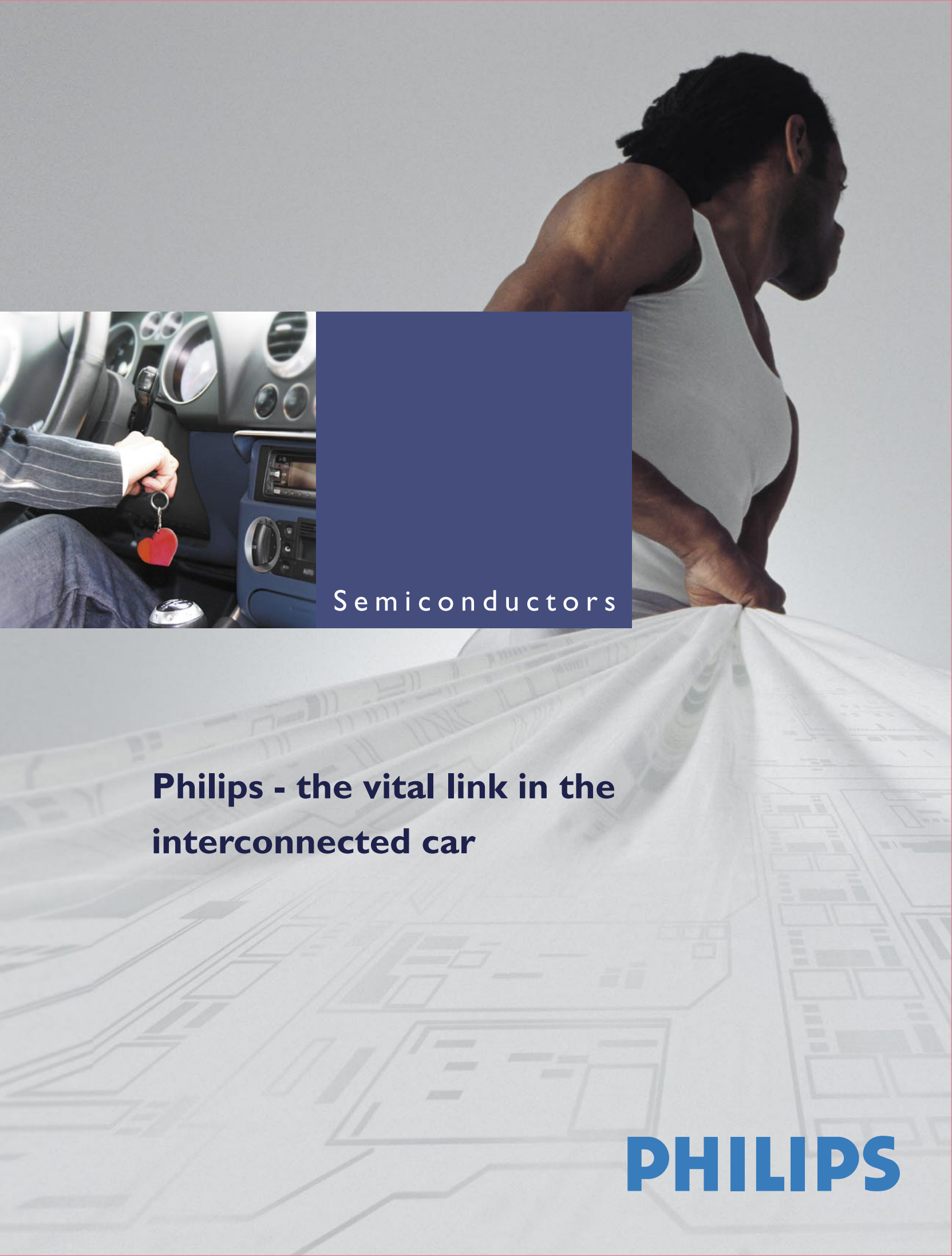




Semiconductors

**Philips - the vital link in the  
interconnected car**

**PHILIPS**



## Advancing silicon for automotive processes

Philips' SOI (Silicon-on-Insulator) technology is the foundation for the outstanding performance of our IVN solutions. Developed specifically to integrate different device types - power, analog and digital - on a single die, Philips' A-BCD (Advanced Bipolar-CMOS-DMOS) family of fabrication processes brings crucial protection to sensitive electronics, low power and superior EMC performance. This effectively eradicates potentially perilous miscommunication in the electrically noisy and hazardous automotive environment.

## Connecting flexible intelligence in the car

Meeting the comfort, convenience, safety and security demands of automotive consumers, as well as increasing driving pleasure and minimizing environmental impact, continues to extend the scope of in-car electronic systems. One of the key enabling technologies behind this trend is In-Vehicle Networking (IVN) and, acknowledged by the industry as the leading innovator in this area, Philips provides proven solutions that set the standard.

Our technologies play a vital role in making connected in-car systems a reality. Using proven protocols we've created comprehensive IVN solutions for systems controlling aspects such as traction, ABS, lighting, windows, mirrors and ride comfort. Extensive system research and development at our Automotive Innovation Center (AIC) in Hamburg has given Philips the experience to provide you with leading-edge CAN and LIN solutions, with exceptional EMC performance, low power consumption and system fail-safe features to meet today's and tomorrow's network requirements. And as you would expect, all these solutions meet the stringent automotive quality and service requirements, clearly demonstrated by the shipment in 2004 of Philips' 700 millionth IVN transceiver.

And with Philips as a long-term IVN development partner, you are assured of early access to emerging technologies for steer-by-wire, brake-by-wire and safe-by-wire. Through close working relationships with major automotive customers and core membership of key industry consortia, such as FlexRay and Safe-by-Wire, we are driving forward future automotive system solutions. Together we can transform today's car into the intelligent personal transport system of tomorrow.





## Driving standards

**Philips continues to develop products that are not only compatible with today's systems, but offer designers the flexibility to incorporate the technology of tomorrow, bringing cutting-edge systems to market first. Our CAN/ LIN-bus and advanced SOI silicon process developments have ensured Philips is leading automotive IVN progress. We are ready for 42 V PowerNet, have improved temperature capabilities (up to 175°C) and our involvement in key industry consortia keeps us one step ahead of the competition, bringing you the benefits of the latest developments.**

### CAN

Effectively the standard for powertrain and body electronics networks, CAN-based IVNs are the first step on the path to embedding intelligence throughout the car. From the introduction of the PCA82C250 nearly a decade ago, Philips single-wire, high-speed and fault-tolerant CAN solutions continue to ensure the highest levels of data integrity and system performance. As the leading CAN innovator, our comprehensive family of devices continues to evolve to include capabilities such as low power consumption. And among the many new developments is a family of System Basis Chips (SBCs), which increase integration, reduce cost, simplify design-in and offer a comprehensive set of fail-safe features. These SBCs will help meet the changing needs of the automotive industry, such as the increased use of permanently powered nodes even under 'ignition-off' conditions.

### LIN

Improving both the cost-effectiveness and flexibility of in-vehicle networking, LIN (Local Interconnect Network) can be used to connect various nodes in a localized area, for example the door lock, mirror and window lift in the doors. As an Associate Member of the LIN consortium, Philips has given momentum to the introduction of this latest standard for in-vehicle sub-networks and is driving the re-definition of the LIN physical layer. We're already building on our LIN capabilities to bring

you extended functionality and further reduce system cost through dedicated LIN SBCs and a family of one-chip LIN-slave-node products.

### FlexRay

As the number of sensors, actuators and control systems for safety, reliability and comfort systems within vehicles increases, a time-triggered, high data-rate communication system is essential. Philips is a core founding member of the FlexRay industry consortium, which is developing the communications protocol and physical layer that will form the core of all critical in-car control applications. Drawing on our extensive IVN knowledge, Philips Semiconductors will serve this emerging market, giving our customers early access to dedicated FlexRay chipsets.

### Safe-by-Wire

Driver and passenger safety have always been critical automotive concerns, and manufacturers are already incorporating more and more safety features across their model ranges. As a founding core member of the Safe-by-Wire consortium, Philips is working to develop a rapid and reliable communication system to ensure the right safety function is activated at the right time. This system will simplify passenger restraint system design, allowing rapid customization and reduced costs, so that the highest level of protection is available in all vehicles.



## FlexRay

- TJA1080: high Speed Time Triggered Transceiver
- High Speed Time Triggered Transceiver with Bus Guardian

## CAN

### Single-Wire CAN:

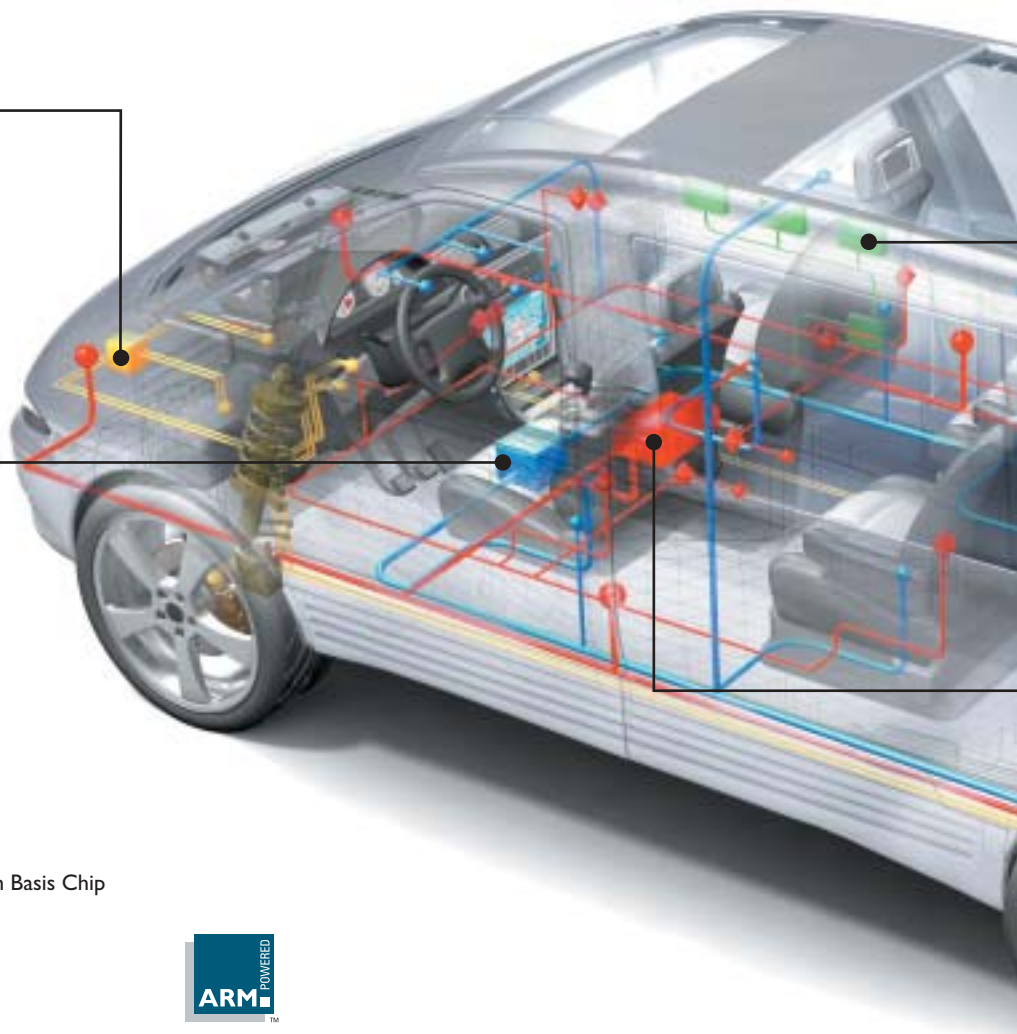
- AU5790

### High Speed CAN:

- PCA82C250: (with standby)
- PCA82C251: (with standby)
- TJA1050: (no standby)
- TJA1040: (with standby)
- TJA1041: (with standby and sleep)
- TJA1041A: (with standby and sleep)
- UJA1065: High Speed CAN / LIN System Basis Chip
- UJA1066: High Speed CAN System Basis Chip

### Fault Tolerant CAN:

- TJA1054: (with standby and sleep)
- TJA1054A: (with standby and sleep)
- UJA1061: Fault Tolerant CAN / LIN System Basis Chip



32-bit ARM-based CAN/LIN Gateway Controller  
• SJA2020





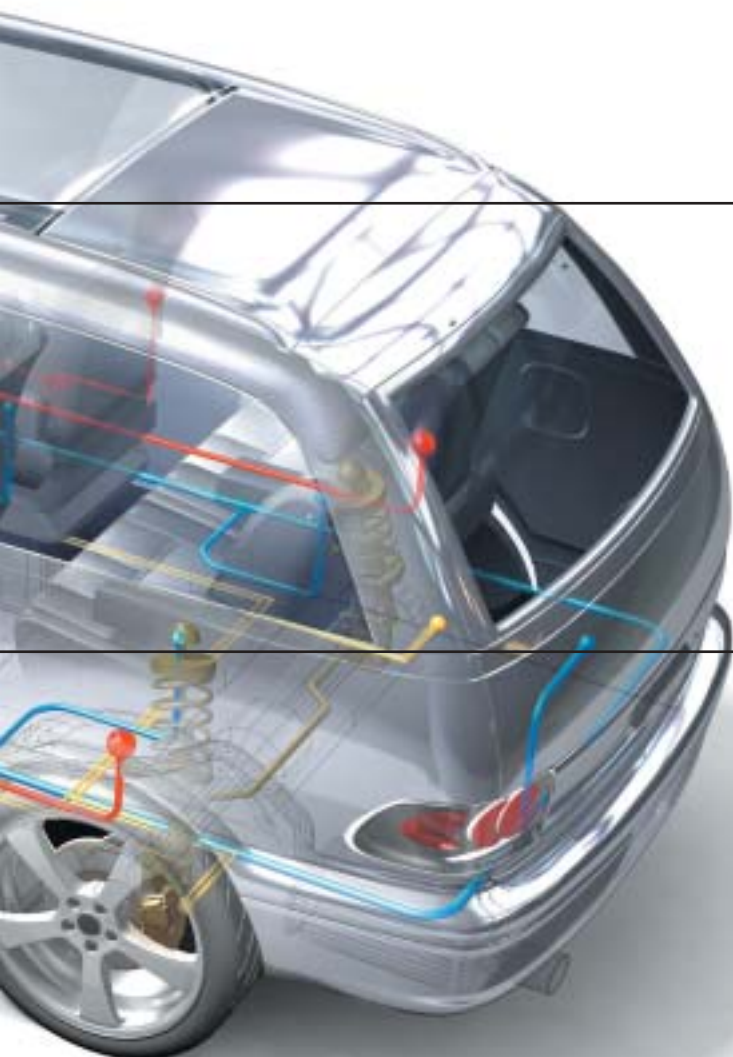
## LIN

- TJA1020: Stand-alone LIN Transceiver
- UJA1061: Fault Tolerant CAN / LIN System Basis Chip
- UJA1065: High Speed CAN / LIN System Basis Chip
- UJA1023: LIN I/O slave
- UJA1069: LIN Slave System Basis Chip



## Safe-by-Wire

- Enabling next generation occupant protection systems
- Advanced open specification network connecting sensors, diagnostics or actuators or all three
- Demonstration available today enabling development of smart occupant safety systems
- Complete silicon solution from Philips
- UJA6402: dual channel master
- UJA6203: raw data slave



Semiconductor



## Building on a leading position

The future of automotive solutions is full of exciting and challenging prospects. Imagine a car that can drive itself and can recognize and adapt to the person behind the wheel. Then picture the driving revolution that would follow such a development - with reduced journey times due to congestion-free traffic systems, greatly improved driver and passenger safety, and pure ease and comfort of driving. This vision may be far-sighted, but the developments taking place today are forming the foundations for future systems to make this a reality.

As the networks within a car increase in complexity, an intelligent system for in-car connectivity is essential. In response to the increasing computation and memory demands that in-car networks and standards such as FlexRay are making, the market is moving towards digital IVN technologies. In parallel with developments extending existing protocols and systems, Philips is developing dedicated digital IVN solutions, leading to faster and more reliable networks, with decentralized intelligence.

With such increasingly complex in-vehicle networks, new and improved gateways, bridges, firewalls and dedicated data link controllers are required to serve the various sub-networks. Philips has close relationships with leading car manufacturers and is working towards incorporating these functions in production cars over the next few years. With our already impressive product range and vast experience and expertise, you cannot find a better partner than Philips to develop the automotive networking solutions of the future.



# Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com). A complete list will be sent to you automatically. You can also visit our website <http://www.semiconductors.philips.com/sales>.

[www.semiconductors.philips.com](http://www.semiconductors.philips.com)

© Koninklijke Philips Electronics N.V. 2004

SCS 76

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

date of release: June 2004  
document order number: 9397 750 11498

Printed in the Netherlands





# A complete CAN family

Philips Semiconductors has played a leading role in establishing CAN as the automotive networking standard. Covering all CAN physical layer options, our product portfolio includes automotive grade high-speed, single-wire and fault-tolerant transceivers along with stand-alone protocol controllers and System Basis Chips. And all our next-generation devices use the flexible SOI technology to offer you best-in-class EMC performance.

## From transceiver to gateway functionality



### Key features

- CAN is a robust protocol – essential for automotive applications
- ISO 11898 and SAE/J2411 are open standards
  - Well documented and fully supported worldwide
- Choice of three CAN physical layer options
  - High-speed (HS) for high data rates
  - Fault-tolerant (FT) for additional robustness
  - Single-wire (SW) for minimum wiring

### Philips CAN portfolio

- Vast experience with CAN in automotive applications
  - Over 700 million transceivers shipped in 2004
- Range of transceivers and protocol controllers for all three CAN physical layer implementations
- A family of innovative products
  - Offer additional fail-safe and low power features
  - Golden devices for new CAN standards
  - Best-in-class EMC performance using versatile SOI technology
- Further integration simplifies individual node and system design
  - Gateway controllers integrating multiple CAN controllers with a 32-bit ARM processor
  - System Basis Chips (SBCs) combining transceivers and voltage regulators with an integral node fail-safe system

The CAN (Controller Area Network) bus is the primary automotive networking protocol for powertrain, backbone bus and body electronics. Involved in its development from the start, Philips has been recognized as the leading CAN innovator ever since our first industry-standard PCA82C250 transceiver set the benchmark for high-speed CAN. We now offer designers a comprehensive portfolio comprising automotive grade transceivers for all CAN physical layer options (high-speed, fault-tolerant and single-wire), stand-alone protocol controllers, System Basis Chips and the latest in IVN gateway ICs, delivering the performance and functionality needed for today's wide variety of in-vehicle networking applications.

As the initial choice of CAN physical layer depends mostly on the network performance required, Philips' CAN solutions ensure your selection meets the highest standards possible. Of the three current implementations of the CAN physical layer, high-speed CAN offers the highest transmission speed (up to 1 Mbits/s) and while fault-tolerant CAN operates at a slower rate (125 kbits/s) it maintains functionality in the case of a broken or shorted bus wire – particularly important in body electronics where the wiring harness is more vulnerable. Both of these CAN physical layer implementations interconnect network nodes via a two-wire twisted pair bus with end-termination. Single-wire CAN is used primarily to reduce wiring in body electronics implementations and for diagnostic purposes, and operates up to 83 kbits/s.

### Delivering value-added performance to automotive CAN networks

The CAN protocol assumes good connections and interference-free signals between nodes in a network. As the CAN protocol itself is not application specific, the automotive environment presents a significant challenge to achieving accurate operation. Obvious risks include physical damage or disturbance and electromagnetic interference. Another important factor is the battery power source, which puts additional demands on power consumption and short-circuit protection. The ISO and SAE CAN standards do not cover all these eventualities, leaving scope for extra functionality and reliability improvements when implementing CAN networks.

# PHILIPS

# A complete CAN family

## From transceiver to gateway functionality



To this end Philips CAN transceiver and controller devices all bring something more to the minimum CAN standard requirements including comprehensive fail-safe features, diagnostics and low-power modes. Many of these additional features are then actually adopted by newer versions of the standards, another process in which Philips' leading role continues. Philips was also the first to introduce automotive grade transceivers to the market.

Failure management in an automotive CAN network serves one goal: keeping the car going. When a failure is detected, the robust CAN protocol handles bus arbitration and the re-transmission of messages, helping minimize effects on the rest of the network and the car. Supplementing the error handling capabilities of the CAN protocol and the selected physical layer, Philips' transceivers all provide additional protective functionality to help safeguard against hardware failures.

Power saving is important for key-off functionality, with networks remaining in standby when the car's engine is switched off. These systems are directly connected to the battery, so power consumption must be kept low. Also, in partial networks (where some nodes remain active when the rest of the network is switched off) the key-on transceivers have to leave the active part of the network unaffected when their power supply is cut. Once again, this is an area where Philips CAN family delivers the added-value features required by today's automotive applications.

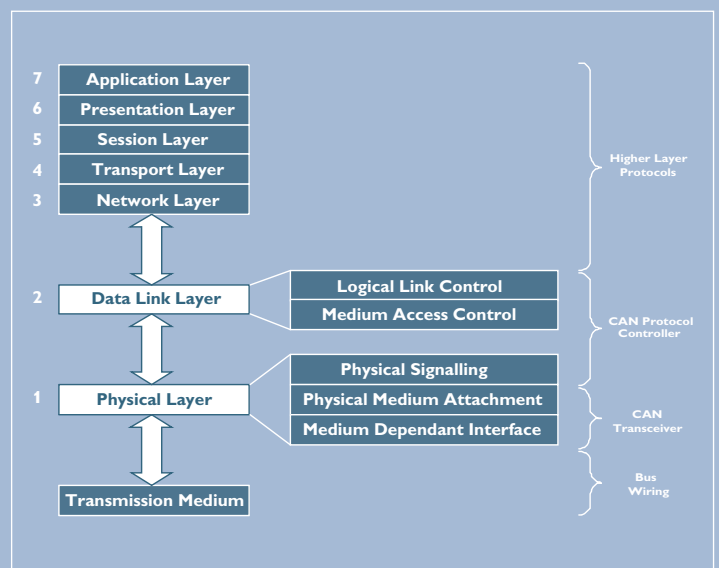


### The CAN protocol

In networked applications the generic ISO/OSI reference model (illustrated) identifies seven distinct communication layers, excluding the actual bus wiring. What is commonly referred to as CAN involves only two of these layers – the data link layer and the physical layer – and is covered by the ISO 11898 standard. In CAN applications the higher-level layers of the ISO/OSI model deal with the application specific processing of the CAN messages. As the actual CAN standard only addresses the basic network communications, this leaves the implementation of features such as fail-safe behaviour and low power modes as proprietary options.

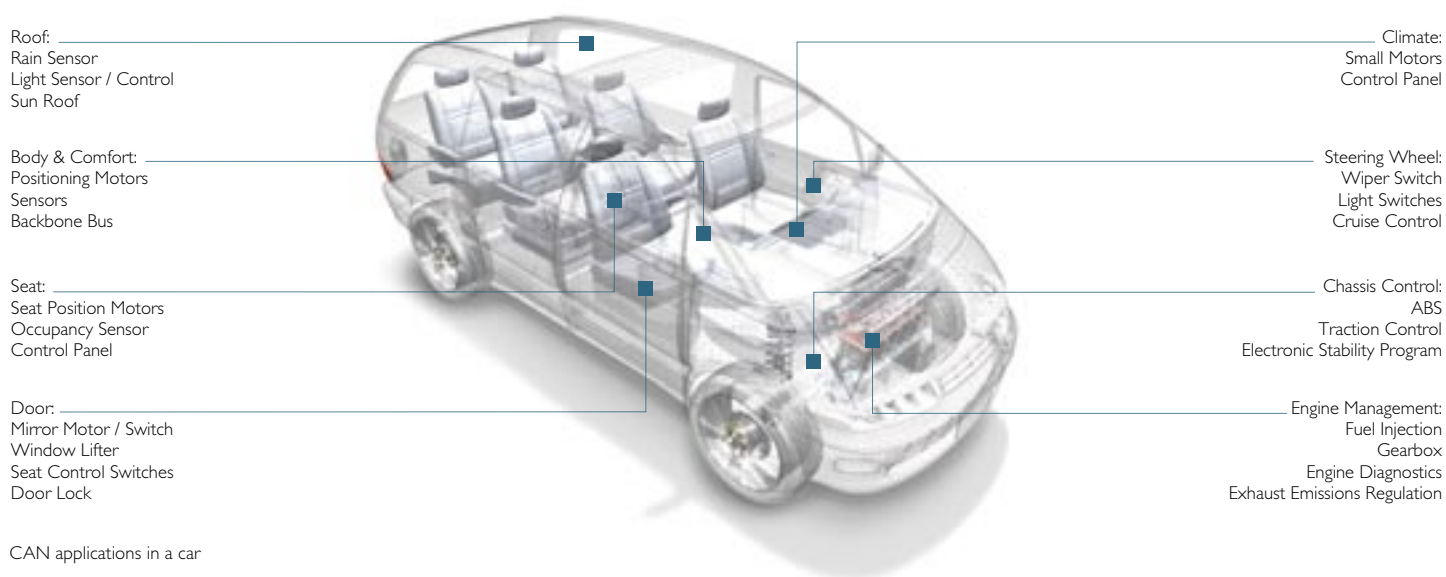
The ISO 11898 standard is divided into three sections: ISO 11898-1 covers the CAN protocol, while parts -2 and -3 handle two of the three standardized implementations of the CAN physical layer (HS-CAN and FT-CAN respectively). Although the transmission medium is not included in the standard, ISO does assume CAN\_L and CAN\_H bus connections and defines the electrical signal levels for them.

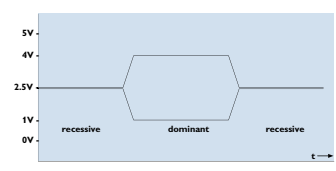
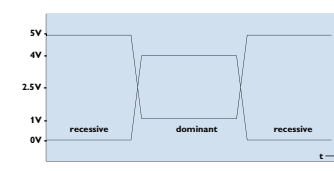
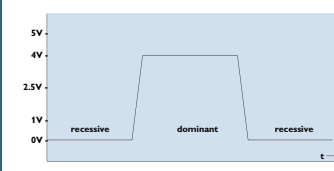
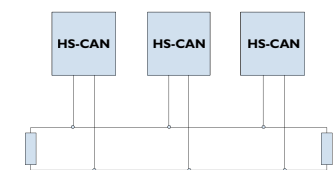
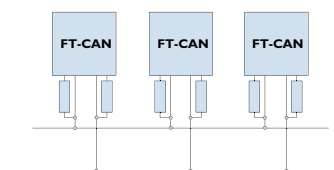
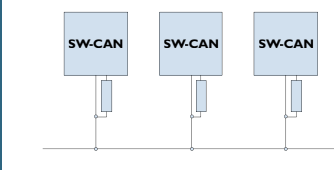
Single-wire CAN is the third standardized CAN physical layer implementation in use today and is covered under the SAE/J2411 standard. SW-CAN uses only the CAN\_H connection, taking the node's ground as reference level. To maintain low electromagnetic emission, the maximum communication speed of SW-CAN systems is limited.



Layer diagram of ISO/OSI reference model

# Overview of CAN physical layer characteristics and application areas



| Features                                       | HS-CAN                                                                                                                                                                                                                                                                                   | FT-CAN                                                                                                                                                                                                                                                                                   | SW-CAN                                                                                                                                                                           |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data link layer standard                       | ISO 11898-1                                                                                                                                                                                                                                                                              | ISO 11898-1                                                                                                                                                                                                                                                                              | ISO 11898-1                                                                                                                                                                      |
| Physical layer standard                        | ISO 11898-2                                                                                                                                                                                                                                                                              | ISO 11898-3                                                                                                                                                                                                                                                                              | SAE/J2411                                                                                                                                                                        |
| Number of bus wires                            | 2 (twisted pair)                                                                                                                                                                                                                                                                         | 2 (twisted pair)                                                                                                                                                                                                                                                                         | 1                                                                                                                                                                                |
| Maximum bus speed                              | 1 Mbits/s                                                                                                                                                                                                                                                                                | 125 kbits/s                                                                                                                                                                                                                                                                              | 33/41.6 kbits/s                                                                                                                                                                  |
| Bus communication signal                       |                                                                                                                                                                                                       |                                                                                                                                                                                                      |                                                                                             |
| Bus termination principle                      |                                                                                                                                                                                                       |                                                                                                                                                                                                      |                                                                                             |
| Bus wire short-circuit and interrupt tolerance | limited short-circuit tolerance                                                                                                                                                                                                                                                          | tolerant against any single bus wire short or interrupt                                                                                                                                                                                                                                  | no tolerance                                                                                                                                                                     |
| Philips transceiver features <sup>1</sup>      | <ul style="list-style-type: none"> <li>- bus dominant time-out</li> <li>- bus clamping protection</li> <li>- partial networking support</li> <li>- stand-by and sleep modes</li> <li>- node power management</li> <li>- local and remote wake-up</li> <li>- failure diagnosis</li> </ul> | <ul style="list-style-type: none"> <li>- bus dominant time-out</li> <li>- bus clamping protection</li> <li>- partial networking support</li> <li>- stand-by and sleep modes</li> <li>- node power management</li> <li>- local and remote wake-up</li> <li>- failure diagnosis</li> </ul> | <ul style="list-style-type: none"> <li>- loss of ground protection</li> <li>- 100 kbits/s flash mode</li> <li>- partial networking support</li> <li>- selective sleep</li> </ul> |
| Automotive applications                        | <ul style="list-style-type: none"> <li>- engine management</li> <li>- backbone bus</li> <li>- body &amp; comfort</li> </ul>                                                                                                                                                              | <ul style="list-style-type: none"> <li>- body &amp; comfort</li> </ul>                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>- body &amp; comfort</li> </ul>                                                                                                           |

<sup>1</sup> depending on the transceiver used



# A complete CAN family

## From transceiver to gateway functionality



### Philips complete CAN portfolio

Philips delivers proven CAN solutions that set the standard. As CAN develops further in the car, we enable manufacturers to bring even higher performance to their CAN applications with drop-in upgrades of existing devices as well as innovative new products. The table gives an overview of Philips CAN product portfolio - further details can be found on separate leaflets.

### The transceiver family for all CAN physical layers

Philips PCA82C250 transceiver became the industry standard for high-speed CAN applications and firmly established Philips Semiconductors as the leading CAN innovator. As CAN developed, we continued to lead with drop-in upgrades of existing high-speed, fault-tolerant and single-wire devices. Fabricated using Philips unique SOI technology, all our latest CAN transceivers offer best-in-class EMI and EME performance and comply fully with ISO and SAE standards, managing proper signal levels and message filtering. In addition, low reverse currents mean unpowered nodes do not disturb networks, while comprehensive fail-safe features help safeguard operation.

### System Basis Chips – effortless solutions to complex challenges

As more and more systems in cars are network based, the complexity of in-vehicle networks increases. Designing a reliable system becomes an escalating challenge, especially where network nodes are permanently connected to the battery. One effective way to reduce this design effort is through smart integration of common ECU functionality and fail-safe features. Highly integrated, Philips System Basis Chips achieve exactly that.

By combining transceivers, voltage regulators, programmable window watchdog and a fully embedded fail-safe system on a single chip, SBCs greatly simplify design effort and FMEA for ECUs and networks. The fail-safe system offers excellent protection against network lock-ups and dead batteries. And the uniform SBC family also makes it very easy to exchange physical layers on ECUs, with maximum re-use of ECU hardware and software.

### Managing networks with dedicated protocol and gateway controllers

Supporting our CAN transceiver family, Philips stand-alone CAN protocol controllers can be used within both automotive and general industrial environments. The successor of the PCA82C200, the SJA1000 implements the enhanced PeliCAN operation mode, which fully supports the CAN 2.0B protocol specification.

With the industry trend to further integrate system functions, we also offer automotive grade 32-bit ARM-based microcontrollers with embedded CAN protocol controllers. These devices are tailored to the specific demands of vehicle dynamics as well as body and safety applications.

### Philips CAN product portfolio

| Philips Products                | HS-CAN                                                              | FT-CAN                                       | SW-CAN                          |
|---------------------------------|---------------------------------------------------------------------|----------------------------------------------|---------------------------------|
| CAN protocol controllers        | SJA1000<br>SJA2020 <sup>1</sup>                                     | SJA1000<br>SJA2020 <sup>1</sup>              | SJA1000<br>SJA2020 <sup>1</sup> |
| CAN transceivers                | PCA82C250<br>PCA82C251<br>TJA1050<br>TJA1040<br>TJA1041<br>TJA1041A | TJA1054<br>TJA1054A                          | AU5790                          |
| System Basis Chips <sup>2</sup> | UJA1065 <sup>3</sup><br>UJA1066                                     | UJA1061 <sup>3</sup><br>UJA1062 <sup>4</sup> | <sup>4</sup>                    |

<sup>1</sup> IVN gateway controller with 32-bit ARM7  $\mu$ C, 6 CAN protocol controllers and 4 LIN protocol controllers

<sup>2</sup> CAN transceiver; 2 voltage regulators, programmable window watchdog and fully embedded fail-safe system

<sup>3</sup> Also includes a LIN transceiver

<sup>4</sup> Study phase

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

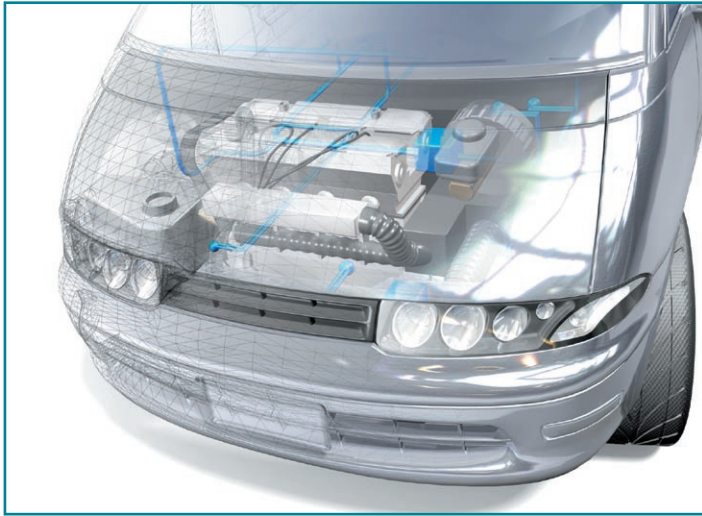
© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



# High-speed CAN transceivers

As the leading supplier of CAN solutions, Philips continues to raise the performance level of this popular in-vehicle networking standard. Our HS-CAN transceiver portfolio addresses a variety of application and network topologies. With advanced functionality such as failure management and power saving, Philips' drop-in upgrades and innovative new products deliver the performance needed by all high-speed CAN applications.



## Key features

- All Philips HS-CAN transceivers are fully ISO 11898-2 compliant
  - Qualified for automotive applications
  - Meet ISO standard over the full voltage and temperature range
- Philips transceivers offer additional features to support the specific needs of automotive HS-CAN applications
  - Fail-safe features and power saving modes help contain network failures and reduce power consumption
- Philips HS-CAN transceivers are found in every type of automotive HS-CAN network in use today
- Philips continues to innovate, based on our vast experience with CAN for automotive applications worldwide

## Philips HS-CAN transceiver portfolio

- PCA82C250 and PCA82C251
  - The most widely used HS-CAN transceivers in the world
- TJA1050
  - Ideal for nodes not requiring a low-power mode
  - Vastly improved EMC and power-off behavior
- TJA1040
  - The best choice transceiver for key-on nodes in partial networks
  - Very low standby current (10  $\mu$ A with wake-up via the bus)
- TJA1041 and TJA1041A
  - The best choice transceivers for continuously powered nodes
  - Very low sleep mode current (20  $\mu$ A for the whole node)

# Enhanced functionality and performance

## Semiconductors

At first only used for engine management networks, High-speed CAN is now commonly used throughout the car, including body and comfort and backbone applications. Addressing the specific requirements of larger and more complicated automotive CAN networks through continuous innovation, functionality such as failure management and low-power modes are now a key component of Philips HS-CAN transceiver family.

Following on from the PCA82C250 Philips offers a choice of stand-alone transceivers fully pin-compatible with their predecessor, that address the additional needs of today's HS-CAN automotive networks. Each provides a combination of power modes and protective features to suit specific applications and network topologies throughout the car. And with our unique SOI process, EMC performance is greatly improved – in some designs completely eliminating the need for choke suppressor coils.

## TJA1041 / 1041A – flagship solutions

Philips' most advanced stand-alone HS-CAN transceivers, both the TJA1041 and TJA1041A feature a sleep mode with inhibit output in addition to a standby mode, providing power supply control for the entire node. This results in a node with extremely low standby current, yet is still capable of waking up via the bus. An extended feature set includes fail-safe features, network diagnostics, local / remote wake-up source recognition and a logic level shifter to interface with microcontrollers between 2.8 and 5 V. Combined, these features make the TJA1041(A) the best choice for permanently connected nodes and partial networks.

## TJA1040 – remote wake-up and 'invisible' nodes

The TJA1040 transceiver offers the functionality of the TJA1050, plus a very low power standby mode with remote wake-up capability via the bus. Bus reverse current is zero when power to the TJA1040 is off, so unpowered nodes do not affect the rest of the network – effectively it is invisible. This makes the TJA1040 ideal for the key-on nodes in partial networks and being fully pin-compatible with the PCA82C250 / 251 and TJA1050, provides a simple migration path for automotive designers.

# PHILIPS

# High-speed CAN transceivers

## Enhanced functionality and performance



### TJA1050 – the efficient HS-CAN transceiver

Compared to the PCA82C250 / 251, the TJA1050 operates with very low bus reverse currents, ensuring unpowered nodes have a minimal effect on the rest of the network. Fail-safe features also prevent nodes from locking into a dominant state and blocking network communication. The TJA1050 helps simplify design-in and reduces overall system costs, allowing manufacturers to offer enhanced functionality in lower model ranges. It is particularly suited to key-on applications or nodes, where a standby mode is not needed.

### PCA82C250 / 251 – popular worldwide standard

Philips continues to build on the success of the PCA82C250, which set the worldwide standard for automotive grade HS-CAN transceivers. With over 250 million units sold it is the most widely used HS-CAN transceiver and is still an excellent choice for key-on HS-CAN networks today, as is the PCA82C251 – a 24 V version for trucks and buses. Originally intended for engine management networks, these devices are now found in every type of in-vehicle HS-CAN network.

### Portfolio overview

|                                                 | PCA82C250 / 251                                               | TJA1050                                     | TJA1040                                                   | TJA1041 / 1041A                                                |
|-------------------------------------------------|---------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| ISO 11898-2 compliant                           | Yes                                                           | Yes                                         | Yes                                                       | Yes                                                            |
| Standby mode with remote wake-up                | Yes                                                           | No                                          | Yes                                                       | Yes                                                            |
| Sleep mode with remote wake-up                  | No                                                            | No                                          | No                                                        | Yes <sup>1</sup>                                               |
| Local wake-up                                   | Yes                                                           | -                                           | Yes                                                       | Yes                                                            |
| Power consumption in standby/sleep mode         | 100 $\mu$ A                                                   | (5 mA)                                      | 10 $\mu$ A                                                | 20 $\mu$ A (for the entire node)                               |
| Invisible when unpowered                        | No                                                            | Approx                                      | Yes                                                       | Approx                                                         |
| Listen-only mode                                | No                                                            | Yes                                         | No                                                        | Yes                                                            |
| TXD dominant timer                              | No                                                            | Yes                                         | Yes                                                       | Yes                                                            |
| Enhanced bus clamping protection (RXD failures) | No                                                            | No                                          | No                                                        | Yes                                                            |
| Bus failure diagnostics                         | No                                                            | No                                          | No                                                        | Yes                                                            |
| Overload protected                              | Yes                                                           | Yes                                         | Yes                                                       | Yes                                                            |
| Split termination output pin                    | No                                                            | No                                          | Yes                                                       | Yes                                                            |
| $\mu$ C compatibility                           | 5 V                                                           | 3.0 V to 5 V <sup>2</sup>                   | 3.0 V to 5 V <sup>2</sup>                                 | 2.8 V to 5 V                                                   |
| Package options                                 | DIL8, SO8, bare die                                           | SO8, HVSON8, bare die                       | SO8, bare die                                             | SO14, bare die                                                 |
| Main application areas                          | Key-on nodes, or in permanently powered networks <sup>3</sup> | Key-on nodes in larger networks, no standby | Partial network key-on nodes or permanently powered nodes | Permanently powered nodes and in partial networks <sup>4</sup> |

<sup>1</sup> TJA1041 wakes up with the first positive dominant state on the bus, TJA1041A wakes up after two dominant cycles on the bus

<sup>2</sup> The  $\mu$ C needs to be tolerant of 5 V input voltages (RXD).

<sup>3</sup> Not recommended for key-on nodes in partial networks, to avoid bus loading issues.

<sup>4</sup> TJA1041A is preferred in large networks, for extra robustness against unwanted wake-ups.

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

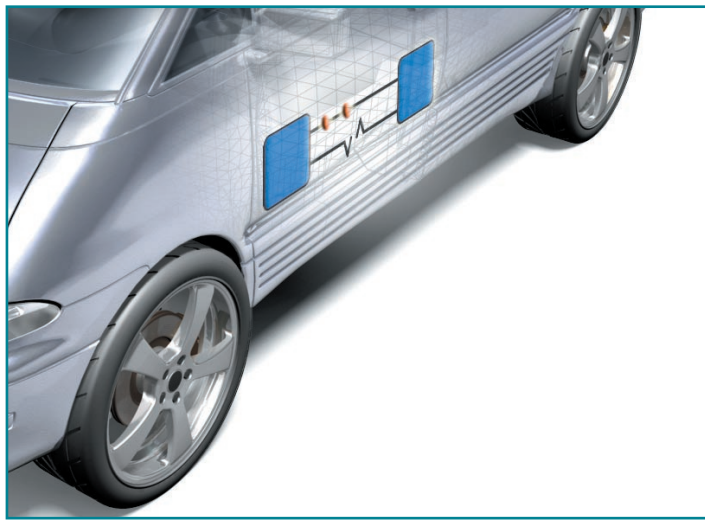
### © Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



# Fault-tolerant CAN product family

As the major proponent of CAN solutions since this networking standard was introduced, Philips continues its industry-leading strategem with innovative developments. Fully ISO 11898-3 compliant, our fault-tolerant FT-CAN transceivers now carry the added benefits of our robust SOI process technology to offer excellent EMC performance.



## Key features

- TJA1054 – ‘golden device’ for defining ISO 11898-3 (FT-CAN) standard
- Excellent EMC performance, both EMI and EME
- Enhanced bus failure management
- Low power mode, with remote and local wake-up capability (also during bus failures)
- Supports unshielded bus wires and chokeless networks
- Integrated temperature protection
- Slope control and receive filtering
- TXD dominant time-out function
- Improved error signaling and behavior during ‘power-loss’ situations
- Passive behavior while unpowered
- Baud rate up to 125 kbaud
- Up to 32 nodes can be connected
- SO14 package and bare die options
- Battery supply can be as low as 5 V

## Additional features of TJA1054A

- 100% compatible with TJA1054
- Improved ESD performance ( $\pm 4$  kV Human Body Model)
- C&S certified for ISO 11898-3 compliance

## Proven performance in FT-CAN transceivers

### Semiconductors

The de-facto body network standard in Europe, 125 kBaud Fault-Tolerant (FT) CAN physical layer maintains functionality in the case of a broken or shorted bus wire. This is of great importance in areas where such failures can occur, particularly seat, door, and roof nodes which are especially vulnerable. If such a fault condition occurs the FT-CAN node simply switches to single-wire operation. Automatic detection of bus system recovery then automatically returns the transceiver to differential mode when the failure has been cleared.

Philips has been involved in the development of FT-CAN transceivers since the inception of this physical layer and continues to introduce enhanced performance and improved functionality. The original PCA82C252 has been succeeded first by the TJA1053, then the TJA1054 and more recently with the TJA1054A, which features advanced Bus Failure Management functionality and improved EMC performance through the use of Silicon-On-Insulator (SOI) technology. In fact, during the definition of the ISO 11898-3 FT-CAN standard, the TJA1054 was used as the ‘golden device’ and became the de-facto FT-CAN transceiver standard due to its excellent performance on the road.

Built using Philips proven SOI mixed-signal process technology, the TJA1054(A)’s best-in-class EMI and EME performance supports unshielded bus wires and chokeless network designs, reducing overall manufacturing costs. The TJA1054 also offers a ‘Low Power’ mode achieving only 30  $\mu$ A power consumption in sleep mode, while the TJA1054(A) includes wake-up capability via the bus lines as well as local wake-up. With the TJA1054 setting the standard in FT-CAN transceivers, the TJA1054A enhances performance still further by offering a 4 kV human body model, thereby doubling the ESD performance.

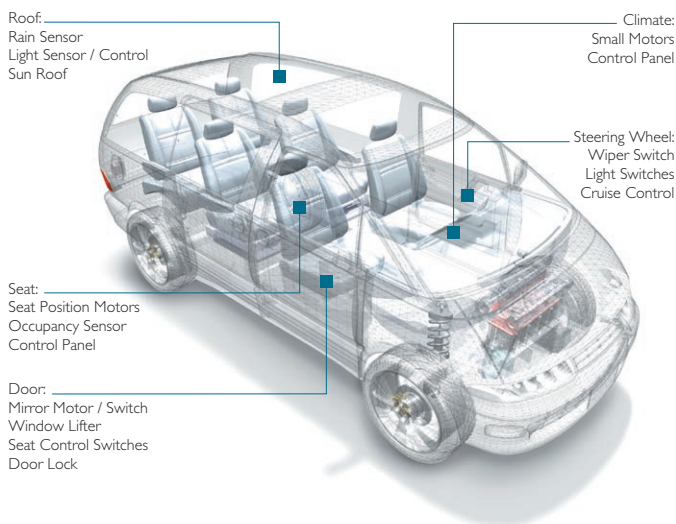
Next generation FT-CAN transceivers will further improve ESD performance and enhancements in transceiver performance will be achieved with new releases of Philips SOI process.

# PHILIPS

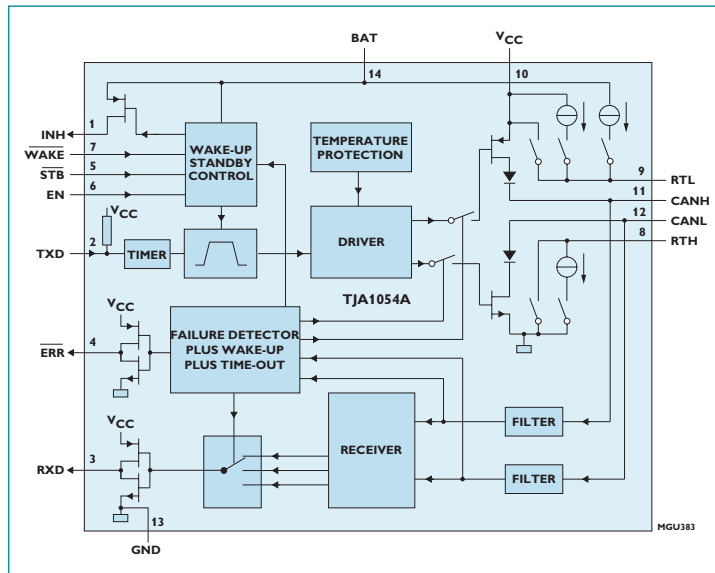


# Fault-tolerant CAN product family

Proven performance in FT-CAN transceivers



FT-CAN applications



Block diagram of TJA1054A

## Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

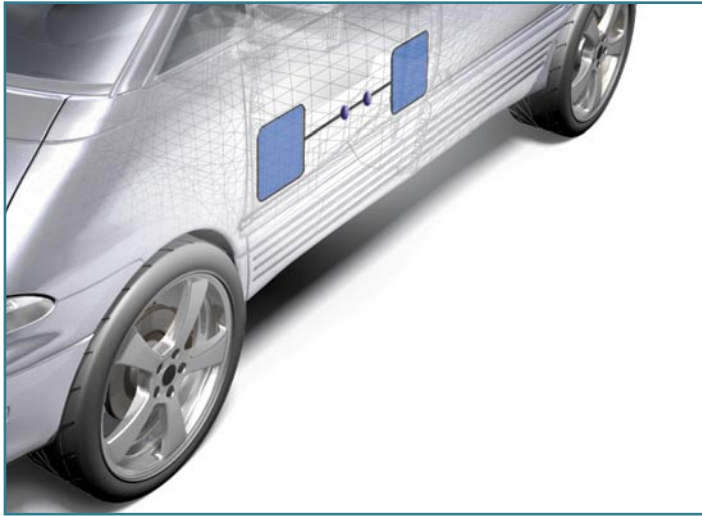


Date of release: June 2004  
Document order number: 9397 750 13039

Published in The Netherlands

# AU5790 single-wire CAN transceiver

Philips family of single-wire controller area network (SW-CAN) transceivers is designed for use in cost-sensitive vehicle body networking applications. Compared with current alternative systems, the AU5790 reduces wiring, assembly and maintenance costs for automotive manufacturers.



## Key features

- Supports in-vehicle class B multiplexing via single bus line with ground return
- 33 kbits/s CAN bus speed with loading as per J2411
- 83 kbits/s high-speed transmission mode
- Low RFI due to output wave shaping
- Direct battery operation with protection against load dump, jump start and transients
- Bus terminal protected against short-circuits and voltage transients
- Thermal overload and loss of ground protections
- Supports communication between control units even when network in low-power state
- Advanced sleep / wake-up functionality
- 70  $\mu$ A typical power consumption in sleep mode
- 8- and 14-pin small outline packages
- $\pm 8$  kV ESD protection on bus and battery pins
- Low number of external components

## Key applications

- Body electronics
- Comfort systems

## Reducing costs for vehicle body networks

### Semiconductors

Benefiting from features such as high-speed mode, embedded protections and advanced functionality to minimize power consumption, the AU5790 SW-CAN device delivers top-class performance. Primarily intended for in-vehicle multiplex and dedicated network applications, it provides an interface between the CAN data link controller and a single wire physical bus line in a range of applications, such as climate control, door locks, instruments clusters, seat positioning and other body and convenience systems.

The AU5790 provides advanced sleep/wake-up functions, minimizing power consumption and reducing battery load when a vehicle is parked, allowing enhanced battery life during ignition-off operation.

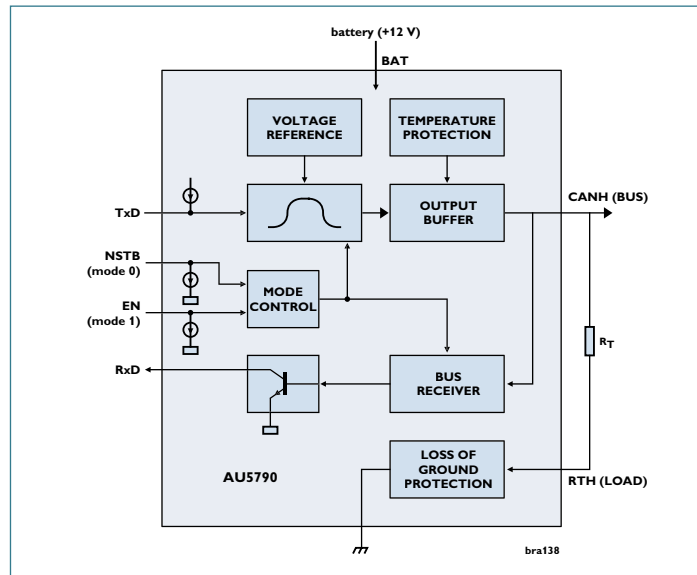
In addition, the unique high-voltage wake-up feature allows normal bus communications without disturbing any sleeping nodes. A bus speed of up to 33.3 kbits/s can be achieved on a network with 32 full bus nodes, although higher speeds are possible with modified bus loading. Fast transfer of larger blocks of data is supported using the 83 kbits/s high-speed data transmission mode.

Adding to Philips Semiconductors' extensive automotive product portfolio, the AU5790 product family offers automotive manufacturers a fourfold increase in performance compared with single-wire non-CAN body systems. In addition, Philips Semiconductors' new family of CAN transceivers features built-in fault tolerance for protection against loss of system ground and communications disruptions during these events. The new transceiver family includes robustness against electrostatic discharge by featuring 8 kV ESD capability on the battery and CanH pins and has built-in surge protection, ensuring full compliance with SAEJ113 / ISO 7637-1 specifications.

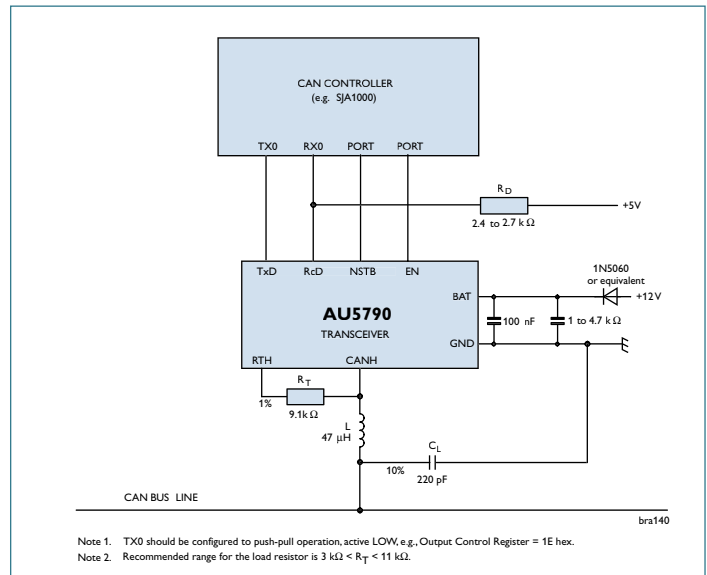
# PHILIPS

# AU5790 single-wire CAN transceiver

## Reducing costs for vehicle body networks



AU5790 block diagram



Application circuit example for the AU5790

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



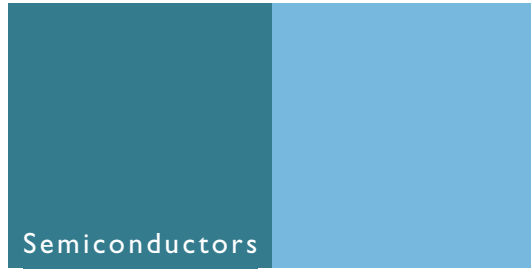
Date of release: June 2004  
Document order number: 9397 750 13071

Published in The Netherlands

# SJA1000

## Versatile stand-alone CAN controller IC

Extending the versatility of its CAN portfolio, Philips' SJA1000 protocol controller provides a stand-alone CAN controller with extended PeliCAN functionality. In combination with Philips range of CAN transceivers this enables flexible implementation of the CAN bus in both industrial and automotive applications.



### Key features

- Pin and electrical compatibility with PCA82C200 stand-alone CAN controller
- BasicCAN (default - PCA82C200) and PeliCAN modes
- Extended receive buffer (64-byte FIFO)
- CAN 2.0B protocol compatibility (extended frame passive in PCA82C200 compatibility mode)
- Supports 11-bit and 29-bit identifiers
- Bit rates up to 1 Mbits/s
- 24 MHz clock frequency
- Interfaces to a variety of microprocessors
- Programmable CAN output driver configuration
- Extended ambient temperature range (-40 to +125 °C)
- Fully automotive qualified

CAN (Controller Area Network) was originally developed for the automotive market, but its performance attributes also make it an ideal solution for industrial environments. The SJA1000 is a stand-alone controller for the CAN bus. As the successor to Philips' PCA82C200 controller (BasicCAN), the SJA1000 offers even greater performance and versatility for all CAN applications.

The SJA1000 includes the enhanced PeliCAN mode of operation, which supports the CAN 2.0B protocol specification and several new features. Offering extended frame messages (29-bits, compared to the 11-bits of the 2.0A protocol), the CAN 2.0B protocol provides over 536 million unique identifiers.

### PeliCAN mode extensions

- Error counters with read/write access
- Programmable error warning limit
- Last error code register
- Error interrupt for each CAN-bus error
- Arbitration lost interrupt with detailed bit position
- Single-shot transmission (no re-transmission)
- Listen only mode (no acknowledge, no active error flags)
- Hot plugging support (software driven bit rate detection)
- Acceptance filter extension (4-byte code, 4-byte mask)
- Reception of 'own' messages (self reception request)

An extensive range of Philips microcontrollers and transceivers complement the SJA1000, ensuring designers can find the best solution for their needs.

# PHILIPS

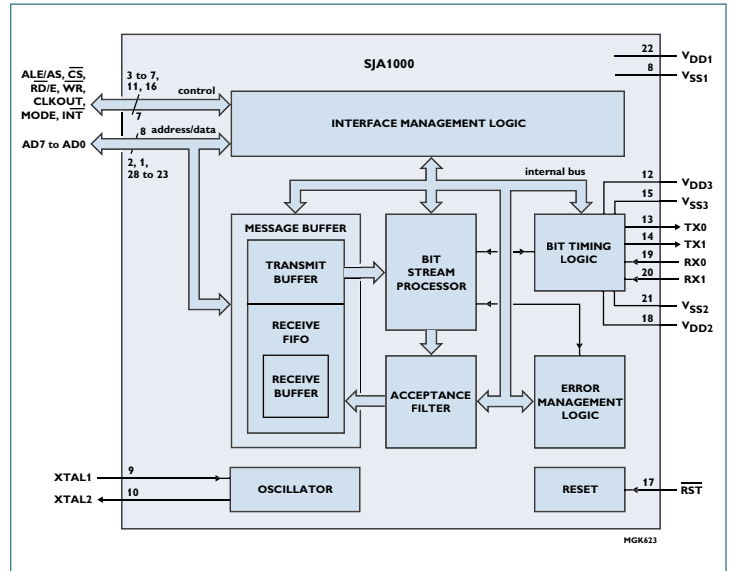
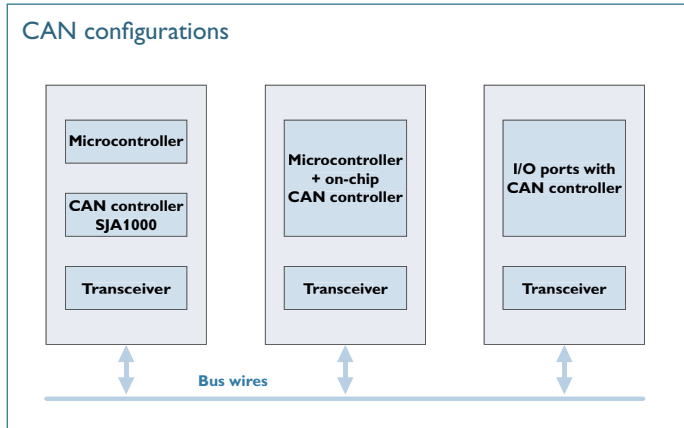


# SJA1000

## Versatile stand-alone CAN controller IC



### CAN configurations



SJA1000 block diagram

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

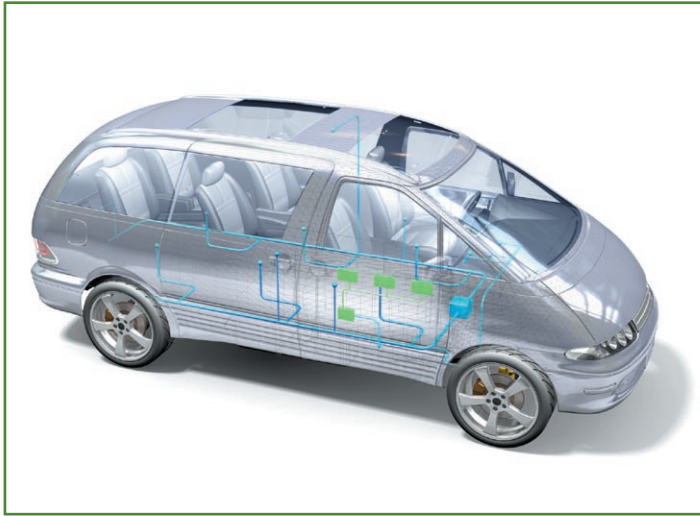


Date of release: June 2004  
Document order number: 9397 750 12954

Published in The Netherlands

# A complete LIN portfolio

A recognized leader in in-vehicle networking (IVN), Philips Semiconductors is a key player in promoting the Local Interconnect Network (LIN) bus as the industry standard for low-cost communications in automotive applications. We offer an extensive range of LIN solutions giving customers the utmost freedom and flexibility when designing LIN based automotive systems.



## Key features

- Single-wire bus
- Single-master / multiple-slave concept
- Based on standard UART / SCI serial data link
- Maximum transmission speed of 20 Kbit/s
- Self-synchronization of slave with RC-based clock source

## Key benefits

- Low cost
- Shorter time-to-market
- Reduced design risk through guaranteed interoperability
- High system reliability
- Enhanced diagnosis capability
- Distributed power
- Frees up backbone bandwidth
- High re-use factor leading to reduced design costs
- Easy to use
- Reduced wiring harness



# Top-level solutions for local bus applications



The LIN physical layer is an enhanced version of the ISO 9141 standard offering improved Electromagnetic Emission (EME) and Electromagnetic Immunity (EMI) performance. As a low-cost, low-speed (class-A) serial communication system it provides a cost-effective communication channel for local automotive systems. Based on a standard UART/SCI serial data link interface, LIN is a single-master/multiple-slave concept. As the master is exclusively in control of bus access no bus arbitration is required.

Interconnecting locally applied sensors, actuators and switches (in e.g. switch panels, door and seat modules) to the CAN backbone with a standardized front end, LIN significantly helps reduce costs and design-in effort for a wide range of human reaction time triggered applications, such as powered mirrors and windows in each door for example.

As an Associate Member of the LIN consortium, Philips has given momentum to the introduction of this latest standard for in-vehicle sub-networks and has driven the re-definition of the LIN physical layer. To provide a complete suite of IVN solutions, Philips offers a growing family of LIN 2.0 compatible ICs complementing our proven CAN portfolio – from stand-alone LIN transceivers and both general-purpose and application-specific LIN slaves up to highly integrated CAN/LIN System Basis Chips (SBCs).

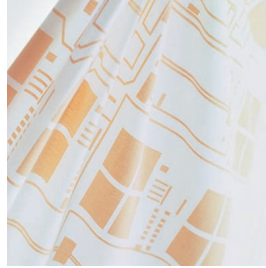
## Stand-alone LIN transceivers

Our stand-alone LIN transceivers deliver the robustness and quality that is essential to automotive network design. For example, the TJA1020 first-generation transceiver exhibited best-in-class performance in independent electromagnetic compatibility (EMC) tests. It also boasts very low power consumption and unique fail-safe features, making it ideal for solid, cost-effective LIN master and slave applications

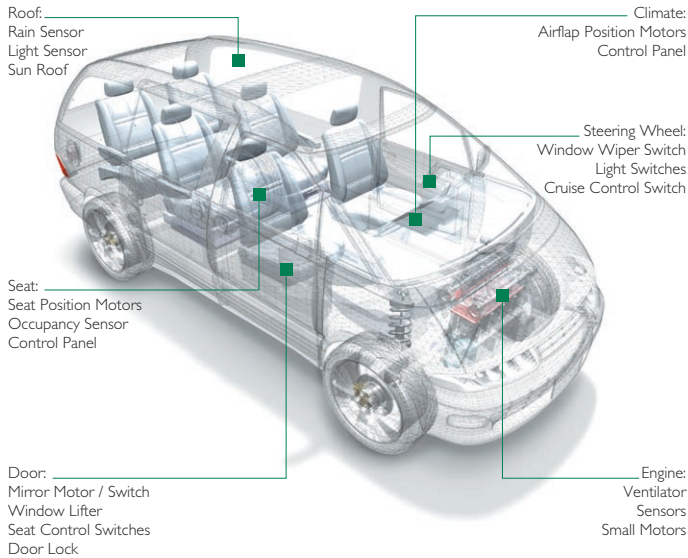
# PHILIPS

# A complete LIN portfolio

## Philips' LIN product portfolio



[www.semiconductors.philips.com](http://www.semiconductors.philips.com)



Integrated LIN Slave Nodes - Typical Target Applications

### LIN slaves

We also offer an extensive choice of general-purpose and application-specific LIN slaves. These LIN slaves, commonly used in local applications such as door, roof and seat assembly units, are fully configurable via the LIN bus in accordance with the LIN 2.0 configuration standard. Besides general I/O functionality, our general-purpose slaves feature internal LIN slave termination, very low quiescent current and a configurable 'limp home' state if required. Meanwhile, our application-specific LIN slaves are single-chip solutions for systems including climate and headlight control. They incorporate a LIN transceiver, protocol handler, dedicated application controller and any necessary power components.

### System Basis Chips

Integrating into a single chip all main peripheral functions around the microcontroller of a typical automotive ECU in a smart approach, Philips SBC family allows customers to re-use hardware and software development to improve time-to-market and save on development and qualification costs. Our LIN-based SBCs include LIN transceiver, voltage regulator, watchdog and embedded fail-safe system, resulting in reliable and cost-effective modules. We offer CAN/LIN SBCs for LIN masters as well as LIN-only SBCs targeting LIN slave implementations using stand-alone microcontrollers.

### Digital IVN: gateway and body controllers

As in-vehicle networks become more complex, Philips is launching a family of 32-bit ARM-based automotive microcontrollers featuring full gateway and / or body control functionality. The first member of this family, the SJA2020 includes four dedicated LIN masters plus six CAN channels. Each LIN master is equipped with receive and transmit message buffers, message interrupts and other features to reduce CPU load.

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com). A complete list will be sent to you automatically. You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

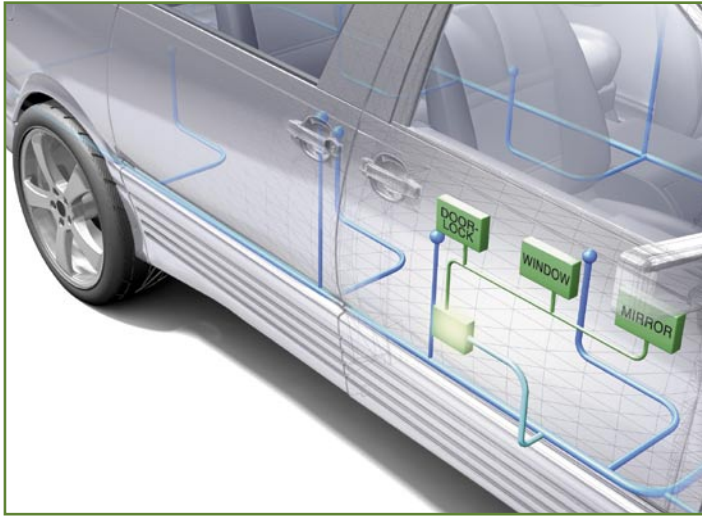


Date of release: June 2004  
Document order number: 9397 750 13131

Published in The Netherlands

# TJA1020 LIN transceiver

Strengthening Philips' leadership in IVN solutions the TJA1020 dedicated LIN transceiver is an essential component in any automotive LIN system. Ideal for solid and cost-effective LIN master and slave node implementations, this first IC in Philips growing LIN family offers best-in-class EMC and power consumption and boasts a range of built-in features.



## Key features

- Baud rate up to 20 kBaud
- Very low Electromagnetic Emission (EME) and high Electromagnetic Immunity (EMI)
- Low-slope mode further reduces EME
- Passive behavior in unpowered state
- Input levels compatible with 3.3 and 5 V devices
- Integrated termination resistor for LIN slave applications
- Wake-up source recognition (local or remote)
- Supports K-line like functions
- 'LIN shorted to ground' protection
- Very low current consumption in sleep mode with local and remote wake-up capability

## Key benefits

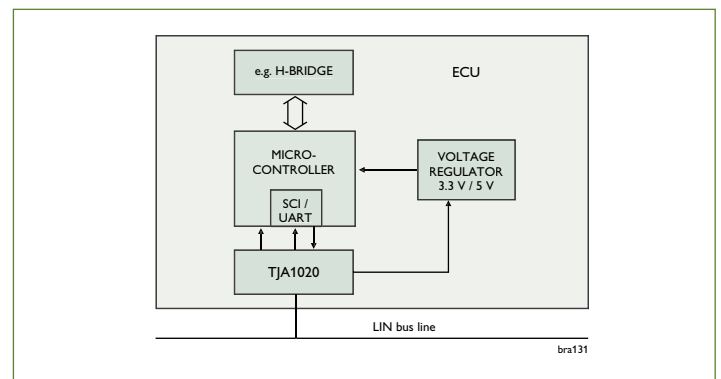
- Capable of directly driving voltage regulator, allowing use of lower-cost regulators
- Best-in-class low power consumption (3  $\mu$ A for entire node)
- Transmit data (TXD) dominant time-out function prevents network lock-up
- Best-in-class Electromagnetic Compatibility (EMC) performance verified by independent tests

# Dedicated stand-alone LIN transceiver

Semiconductors

Philips' TJA1020 interfaces the master / slave protocol controller to the physical bus in a Local Interconnect Network (LIN). It is a general-purpose stand-alone LIN transceiver, ideal for in-car sub-bus applications using baud rates from 2.4 to 20 kbaud. With I/O levels compatible with 3.3 V and 5 V devices, it can be designed-in with a wide range of microcontrollers. On-chip circuitry safeguards against transients while LIN-to-ground short-circuit protection prevents battery discharge. Along with a transmitter-dominant monitor, they ensure the TJA1020 is completely fail-safe. Moreover, the TJA1020 exhibits completely passive behavior in its unpowered state.

Solid automotive network design heavily depends on the robustness of the transceiver. However, a challenging factor of single-wire bus systems like LIN is EMC. The TJA1020 boasts best-in-class EMC performance together with high EMI and extremely low EME, which can be further reduced by using the 'low slope' operating mode. Despite its extreme low current consumption in sleep, the TJA1020 will adequately react on any valid local or remote wake-up event.



Typical LIN ECU

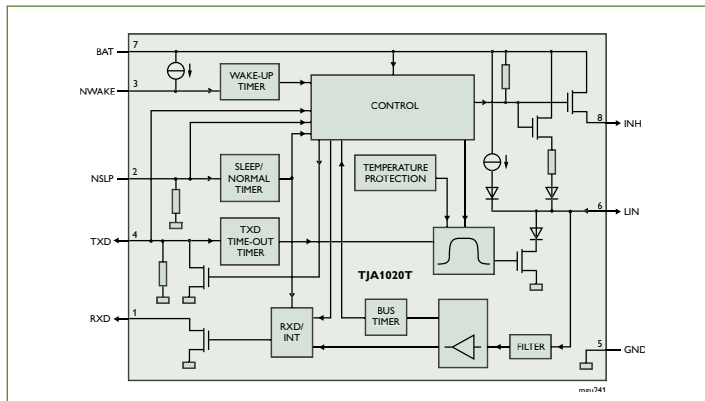


# TJA1020 LIN transceiver

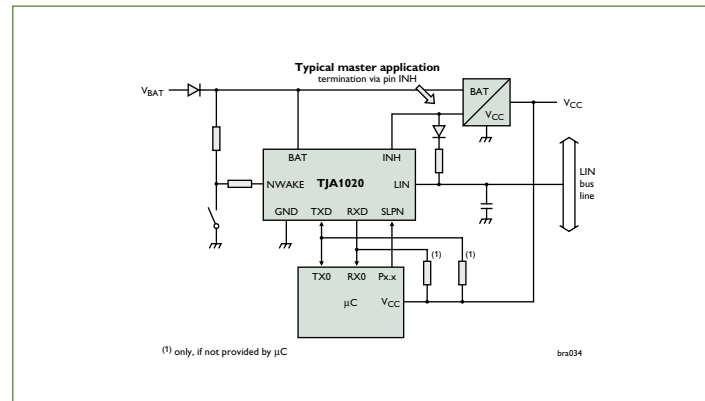
## Dedicated stand-alone LIN transceiver



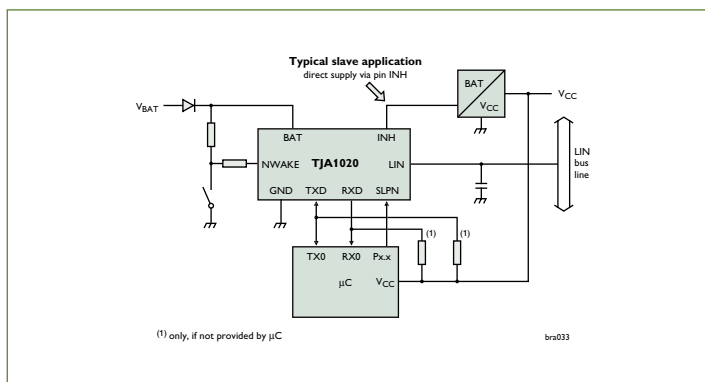
[www.semiconductors.philips.com](http://www.semiconductors.philips.com)



TJA1020 block diagram



Typical LIN master application



Typical LIN slave application

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



Date of release: June 2004  
Document order number: 9397 750 13046

Published in The Netherlands

# UJA1023T LIN I/O slave

An acknowledged leader in in-vehicle networking (IVN) solutions, Philips' latest addition to its dedicated LIN product portfolio is the stand-alone UJA1023T LIN I/O slave. With many on-chip features – including an integrated LIN transceiver – it significantly enhances LIN systems by minimizing application-specific overheads.



## Key features

- Automatic bit-rate synchronization to any bit-rate between 1 and 20 kbit/s
- Integrated LIN transceiver (including 30 k $\Omega$  termination resistor)
- 8 bi-directional I/O pins
- 4x4 switch matrix to support reading a maximum number of 16 switches
- Outputs can be configured as high-and/or low-side driver, cyclic or PWM driver
- Simple 8-bit A/D converter (max. 8 channels)
- Advanced low-power behavior
- On-chip oscillator
- ID configuration via daisy chain or plug coding
- Inputs support local wake-up and edge capturing
- Configurable sleep mode
- Limp home configuration in case of error conditions
- Extremely low emissions
- High immunity against electromagnetic interference
- Bus line protected against transients in an automotive environment
- Extended ambient temperature range (-40 °C to +125 °C)

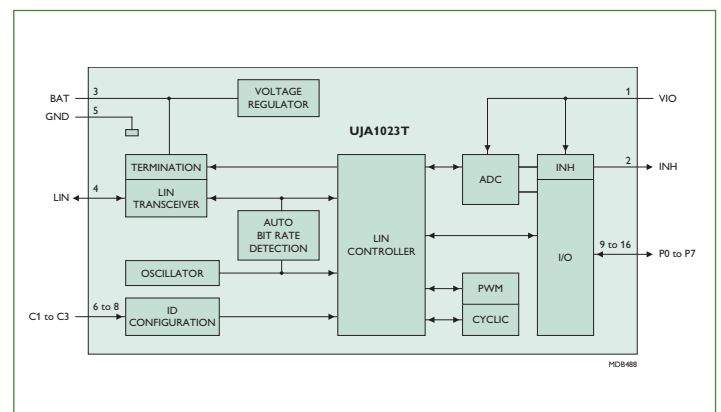
## Versatile, single-chip solution for control networks

### Semiconductors

A fully integrated LIN I/O slave node (LIN physical layer and protocol handling), Philips' UJA1023T delivers a low-cost, single-chip solution for multiple application fields – replacing several components commonly used in control units for input and output handling. It incorporates an automatic bit-rate synchronization circuit, enabling the device to accept any bit-rate between 1 kbit/s and 20 kbit/s.

The on-chip LIN 2.0 transceiver, together with the protocol handler, oscillator and 30 k $\Omega$  termination resistor necessary for LIN slaves, handles the LIN protocol autonomously. ID programming is achieved by a master request and slave response message, combined with daisy chain or plug coding function, and the device also has a limp home configuration in case of error conditions.

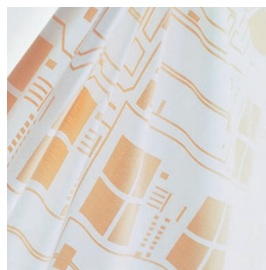
An advanced low-power concept ensures very low power consumption, augmented by an energy-saving, configurable sleep mode that also provides a local wake-up feature. As with all Philips IVN solutions, the UJA1023T offers extremely low emissions and high immunity against electromagnetic interference, while the bus line is protected against automotive transient voltages.



UJA1023T block diagram

# UJA1023T LIN I/O slave

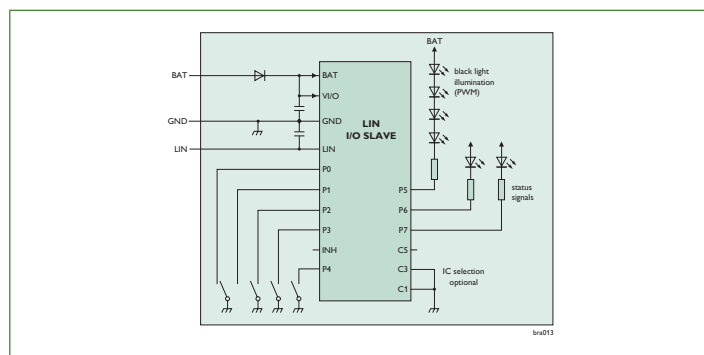
Versatile, single-chip solution for control networks



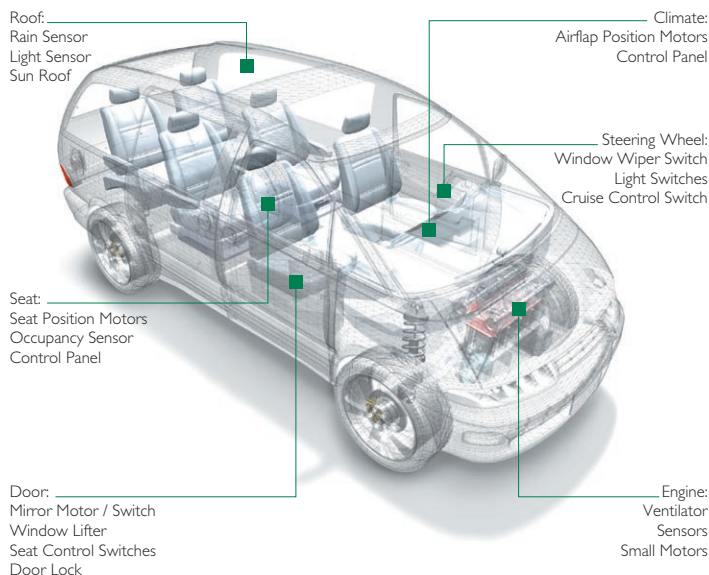
[www.semiconductors.philips.com](http://www.semiconductors.philips.com)

The 8 programmable bi-directional I/O pins can have following functions:

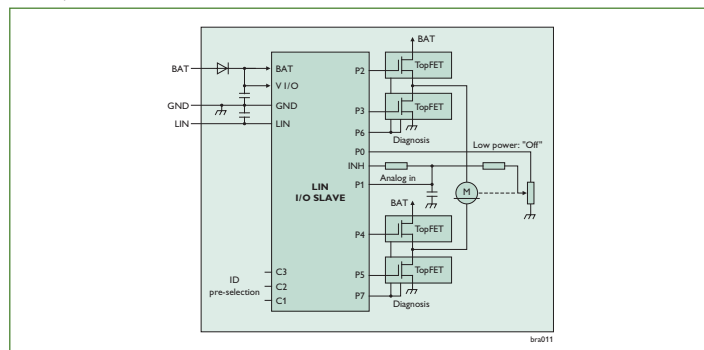
| Input                                                                                                                                                                                                                                   | Output                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Standard input pin</li> <li>Local wake-up in case of sleep mode entered</li> <li>Edge capturing on falling, rising or both edges</li> <li>Analog input pin</li> <li>4x4 switch matrix</li> </ul> | <ul style="list-style-type: none"> <li>Standard output pin as high-side, low-side or push-pull driver</li> <li>Cyclic mode for local wake-up</li> <li>PWM mode, for example for controlling backlight intensity</li> <li>4x4 switch matrix</li> </ul> |



Switch panel with illumination and status feedback



Integrated LIN Slave Nodes - Typical Target Applications



Motor driver with position feed-back

## Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

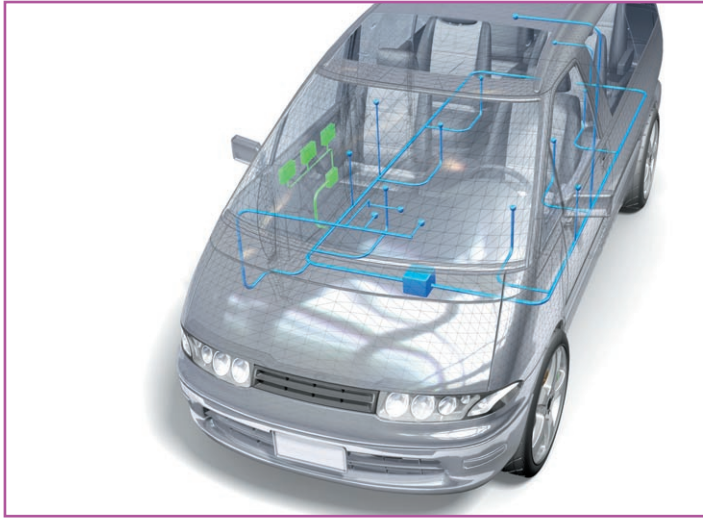


Date of release: June 2004  
Document order number: 9397 750 11956

Published in The Netherlands

# UJA106x System Basis Chips

By smart integration of several components into a single **System Basis Chip (SBC)**, Philips offers you more reliable, smaller and lower-cost **CAN / LIN** network solutions. Safer, quicker and easier to design-in, our **SBCs** reduce component count through extensive on-chip functionality that includes **CAN** and **LIN** transceivers, voltage regulation, watchdogs and fail-safe operation.



## Key features

- Family of SBCs with LIN2.0 compliant and / or ISO 11898 compliant CAN transceivers
- Excellent EMC performance (3rd generation Silicon-On-Insulator process)
- 12V, 24V and 42V system support with low sleep current (typically 50  $\mu$ A)
- Supports 2.5, 3.0, 3.3 and 5V microcontrollers, with automatic interface level
- Programmable window-watchdog with on-chip oscillator, guaranteeing independent fail-safe system supervision
- Cyclic wake-up capability in standby and sleep mode, e.g. for rain sensor application
- Fail-safe coded 16-bit SPI with multiple device capability
- Remote wake-up capability via CAN and LIN buses
- Local wake-up input with current saving feature
- Programmable high-side switch
- Smart operating and power management modes
- Software development mode
- Unique SPI-readable device identification of silicon status and type
- $\pm 8$  kV ESD protection (HBM) and excellent robustness
- Small footprint HTSSOP32 package with low thermal resistance

## Smart integration of CAN/LIN transceiver functionality



Philips System Basis Chips are highly integrated systems-on-chip for IVN systems, addressing the trend for increased network complexity and functionality. Offering the most extensive feature set in the industry, they greatly simplify the design and implementation of ECUs, especially for applications operating under 'ignition-off' conditions. These SBCs combine LIN and /or CAN transceivers, two LDO voltage regulators, an SPI, a watchdog and many diagnostics features on a single chip. They also deliver excellent EMC performance and unique fail-safe system behavior.

## Key benefits

- Lower development cost and risks for both car manufacturers and module makers through unique SBC family re-use approach
- Eases exchange between different physical layer implementations (e.g. fault-tolerant and high-speed)
- Simplifies building of robust and fully fail-safe ECUs for use in 'ignition-off' conditions
- Support a wide range of microcontrollers to maximize flexibility and to lower cost
- Flexibility in network design due to partial networking capability
- Reduction of ECU size and ECU sleep current

## Integrated fail-safe and diagnostic features

Philips SBCs incorporate an extensive set of SPI readable diagnostics features allowing rigorous error handling.

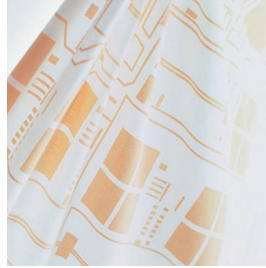
Detection and detailed error reporting on CAN and LIN bus failures (e.g. shorts to GND/BAT, open bus wires etc.) ensure any failure can be easily handled, while TxD dominant and RxD recessive clamping as well as RxD to TxD short detection prevent bus deadlocks. Features such as ECU two-level ground-shift detection, over-temperature warning, ECU battery monitoring (for example to detect chattering battery contact or to save data before microcontroller power down) and signaling of potential RAM-retention errors due to low microcontroller Vcc, all help ensure robust system operation.

# PHILIPS



# UJA106x System Basis Chips

## Smart integration of CAN/LIN transceiver functionality



[www.semiconductors.philips.com](http://www.semiconductors.philips.com)

Other features include guaranteed ramp down of the microcontroller voltage in case of a soft restart, detection of cyclic ECU problems and Global Enable pin to control safety critical hardware.

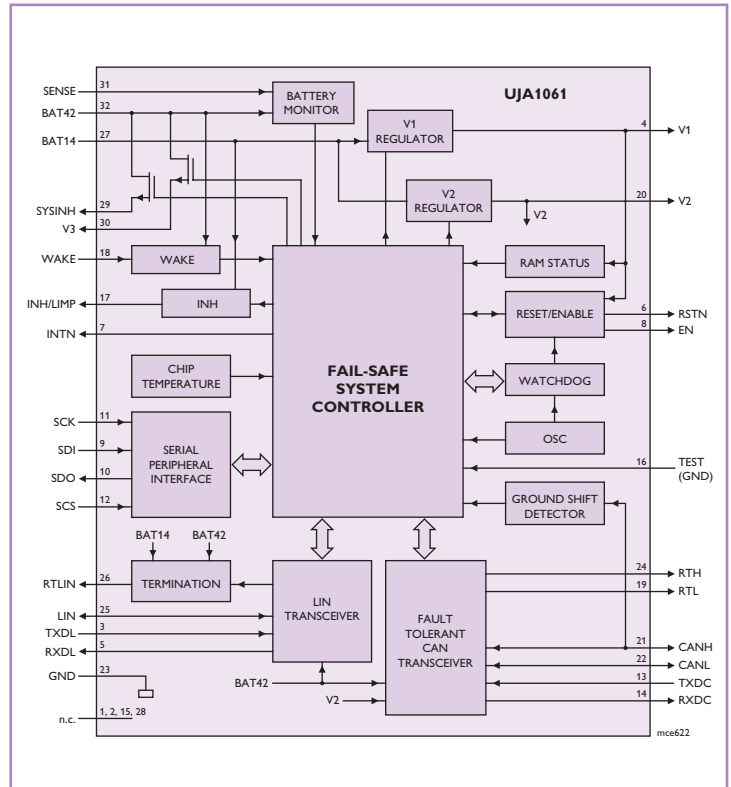
A unique, integrated fail-safe system controller ensures any conceivable local ECU failure stays local, thereby avoiding a global network communication deadlock or an empty battery. If a failure condition occurs the SBC will detect, report and ultimately switch to low power, fail-safe mode ensuring that local failures stay local and do not disrupt the entire bus communication. This is an important factor with the increasing number of bus nodes in today's cars. A 'limp-home' output signal is also included for activating application hardware in case system enters fail-safe state.

### A platform approach for a unique family

All Philips SBCs integrate individual blocks in a smart way, enabling compact solutions delivered in the small HTSSOP32 package: a further benefit for enhanced CAN and LIN networking implementation. Only a microcontroller needs to be added to complete the ECU. By taking a platform approach to Philips SBC, one solution can fit many ECU variants and extending functionality can be achieved by adding more LIN Slaves to the bus. This allows designers to optimize re-use of hardware and software, improving time-to-market and reducing development and qualification costs. In addition, Philips SBC approach maximizes economy-of-scale to support reduction of ECU costs.

### Initial SBC family options

- UJA1061 – ISO 11898-3 compliant fault-tolerant CAN and LIN2.0 compliant SBC
- UJA1065 – ISO 11898-2 compliant high-speed CAN and LIN2.0 compliant SBC
- UJA1062 – ISO 11898-3 compliant fault-tolerant CAN SBC
- UJA1066 – ISO 11898-2 compliant high-speed CAN SBC
- UJA1069 – LIN2.0 compliant SBC



SBC block diagram

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

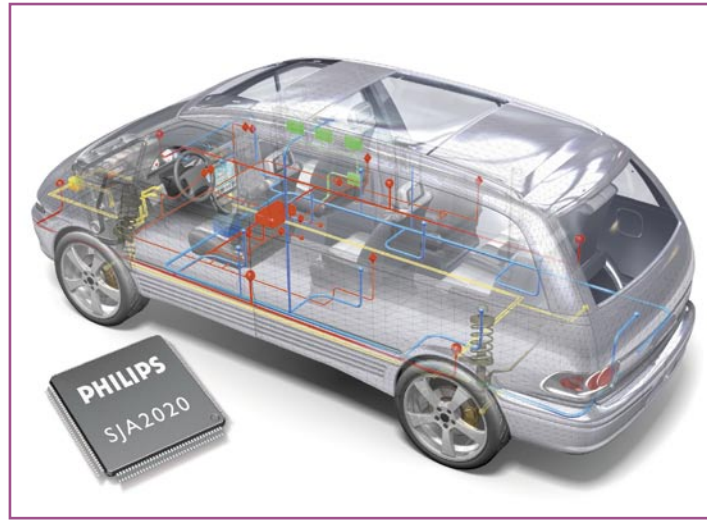
Date of release: June 2004

Document order number: 9397 750 13407

Published in The Netherlands

# SJA2020

The SJA2020 is a 32-bit ARM7-based automotive IVN gateway controller, featuring fast, robust embedded Flash, dedicated in-vehicle network interfaces and a compelling peripheral set in a 144 pin package. With hardware supporting up to 6 CAN and 4 LIN interfaces, the SJA2020 IVN-gateway delivers excellent performance for the emerging digital IVN market.



## Key features

- 32-bit ARM7 microcontroller running at up to 60 MHz
- Fully qualified to AEC Q100 standard for automotive semiconductors
- 256 kB/384 kB embedded Flash with source code protection and 4 Mbit/s fast programming
- On-chip 24 kB static RAM may be used for code and / or data storage
- 6 CAN channels with global acceptance filter
- 4 LIN master controllers implemented in hardware
- 4 industry standard '450' UARTs and one '550' UART with 16 bytes transmit and receive FIFO
- 4-channel 10-bit A/D converter
- Real time clock with on-chip 32 kHz crystal oscillator and dedicated (battery) supply
- Three full-duplex SPI
- 32-bit watchdog with timer change protection
- External 8, 16 or 32-bit bus with four memory banks
- Up to 94 general purpose I/Os
- Ambient operating temperature range from -40 to +105°C
- Dual Supply: CPU 1.8V  $\pm$  5 %, I/O 2.7V to 3.6V, 5V tolerant
- On-board ring oscillator for advanced power saving modes
- LQFP 144 pin package

## 32-bit ARM7 microcontroller with IVN gateway functionality

### Semiconductors

The SJA2020 IVN communication gateway is Philips' first dedicated 32-bit ARM-based automotive microcontroller. Gateway ICs are essential to deal with the additional computing and memory demands created by increasingly complex in-vehicle networks. The SJA2020 has been developed through close relationships with key automotive industry players to ensure it is a cost-effective high-performance solution. Based on our proven embedded core and automotive expertise, it enables optimized receiving, composing, routing and transmission of network messages on different in-vehicle serial bus systems.

### Key benefits

- Efficient CAN message handling with maximum flexibility
- Dedicated LIN message handling frees up CPU processing power for applications
- Extensive feature list allows designers to use the same microcontroller for different applications

### Key applications

- Car gateways with up to 6 CAN and up to 4 LIN networks
- Body control requiring 32-bit CPU performance

Employing Philips' reuse design approach, including all software and hardware gateway functionality, provides designers with the highest level of flexibility in terms of implementation and customization of the gateway. The dedicated gateway hardware supports up to 6 CAN channels and 4 dedicated LIN masters, delivering high-performance gateway functionality with minimal CPU load. A unique Global Acceptance Filter servicing all CAN controllers with a hardware accelerated scanning algorithm ensures extremely short transfer-latency times. All CAN channels are CAN 2.0B compliant with FullCAN receive mode and triple transmit buffers. The 4 dedicated LIN master controllers enable CAN/LIN gateway functionality, while three state-of-the-art SPI interfaces and an external bus interface allow fast access to multimedia networks such as MOST or IDB-1394.

# PHILIPS

# SJA2020

## 32-bit ARM7 microcontroller with IVN gateway functionality



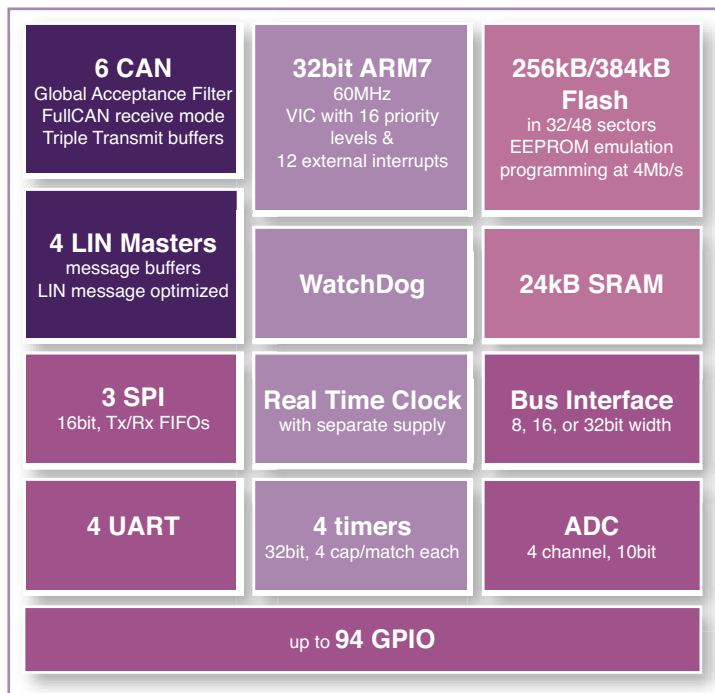
The SJA2020 features a robust 256 kB/384 kB Flash memory system with fast In-System Programming (ISP) via a JTAG interface and In-Application Programming (IAP) via a serial port (e.g. CAN), providing flexibility for data storage field upgrades and EEPROM emulation. In addition with up to 94 general purpose I/Os and 5 UARTs, four 32-bit timers each with 4 capture and match registers and a 4-channel 10-bit ADC, the SJA2020 can also meet a wide variety of automotive application requirements.

An extensive range of OSEK compliant real time operating systems, CAN drivers and embedded software components are available for automotive applications, as are a variety of software development tools provided by several vendors. The SJA2020 starter kit also includes a prototyping board together with demo software drivers.

### ARM7TDMI-S embedded processor Core

The ARM7TDMI-S is a general-purpose 32-bit microprocessor, which offers high performance and very low power consumption. Based on reduced instruction set computer (RISC) principles, the instruction set and related decode mechanism are much simpler than those of micro programmed complex instruction set computers (CISC). This simplicity results in a high instruction throughput and impressive real-time interrupt response from a small and cost-effective processor core.

Clocked by an on-chip PLL that allows CPU operation up to 60 MHz, the 32-bit RISC architecture of the ARM7 core also features a 16-bit thumb mode resulting in higher code density due to a reduced instruction set. The SJA2020 contains a very flexible Vectored Interrupt Controller (VIC) with up to 16 priority levels to interrupt the ARM processor on request. The software development is facilitated by the standard ARM Test/Debug interface with Real-Time In-Circuit Emulator.



SJA2020 block diagram

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com). A complete list will be sent to you automatically. You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



Date of release: June 2004  
Document order number: 9397 750 11975

Published in The Netherlands

# SJA2010

Philips' SJA2010 is a 32-bit ARM7-based microcontroller for automotive body control and gateway applications. Featuring scalable Flash memory, dedicated in-vehicle network interfaces and a compelling peripheral set in a range of package options, the SJA2010 provides designers with the highest level of flexibility for implementing and customizing the on-chip peripherals.



## Features

- 32-bit ARM7, running at up to 60 MHz
- 512 / 768 Kbytes embedded Flash with source code protection and 4 Mbits/s fast programming capability
- 40 Kbytes SRAM
- CAN 2.0B compliant with FullCAN receive mode and triple transmit buffers
- 6 CAN controllers and 8 LIN 2.0 master controllers
- 2 UARTS and 3 full-duplex SPI channels
- Real time clock with on-chip 32kHz crystal oscillator and battery supply
- 4 x 32-bit timers, each with four capture and compare registers
- 18 level-sensitive external interrupt pins
- 32 analog input pins
- 24 PWM outputs, suitable for light dimming and brushless DC motor control
- Up to 160 general purpose I/Os
- Dual supply – CPU 1.8V; I/O 2.7V to 3.6V and 5V tolerant
- Internal programmable ringoscillator for power saving mode
- LQFP144/176/208 and LFBGA208 packages

## 32-bit ARM7 body controller with IVN gateway functionality



The SJA2010 32-bit ARM7 microcontroller includes dedicated hardware for IVN communication, making it the ideal solution for high-performance automotive body control and gateway systems as well as combined body control and gateway applications.

Hardware IP blocks based on Philips' proven CAN experience enable optimized receiving, composing, routing and transmission of network messages on different serial bus systems. A unique 'global acceptance filter' services all CAN controllers with a scanning algorithm, ensuring extremely short transfer-latency times. The SJA2010 supports specific hardware for 6 CAN channels and 8 dedicated LIN 2.0 master controllers, enabling high performance network communication with minimal CPU load.

## Benefits

- Powerful, flexible solution with extensive functionality and I/Os
- Hardware support for CAN and LIN
- Same microcontroller can be used for multiple applications

## Applications

- Body control
- Combined body control / gateways
- Dashboard and comfort systems

Employing Philips' reuse design approach, including all software and hardware gateway functionality, provides designers with the highest level of flexibility in terms of implementation and customization of the gateway. Dedicated PWM hardware blocks make the SJA2010 an ideal solution for typical body control applications such as light dimming and brushless DC motor control. All CAN channels are CAN 2.0B compliant with FullCAN receive mode and triple transmit buffers. Three state-of-the-art SPI interfaces and an external bus interface allow fast access to multimedia networks such as MOST or IDB-1394.

# PHILIPS



# SJA2010

## 32-bit ARM7 body controller with IVN gateway functionality



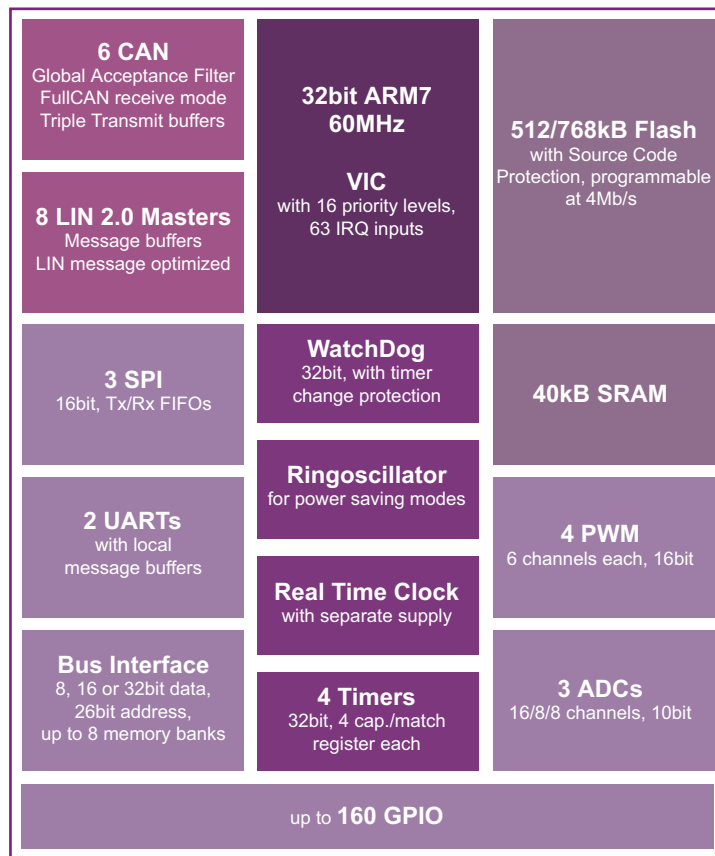
### Universal functionality

The SJA2010 microcontroller is ideal for automotive applications requiring 32-bit processing power and large pin count. Up to 160 general purpose I/O's, 32 analog input pins, 24 PWM outputs together with 3 full-duplex SPI channels, 6 CAN controller and 8 LIN 2.0 master controllers make it a universal solution. The SJA2010 is available in packages with 144 up to 208 pins so that the same microcontroller can be used for different application areas.

### ARM7TDMI-S embedded processor core

The ARM7TDMI-S is a general-purpose 32-bit microprocessor, which offers high performance and very low power consumption. Based on reduced instruction set computer (RISC) principles, the instruction set and related decode mechanism are much simpler than those of micro programmed complex instruction set computers (CISC). This simplicity results in a high instruction throughput and impressive real-time interrupt response from a small and cost-effective processor core.

Clocked by an on-chip PLL that allows CPU operation up to 60 MHz, the 32-bit RISC architecture of the ARM7 core also features a 16-bit thumb mode resulting in higher code density due to a reduced instruction set. The SJA2010 contains a very flexible Vectored Interrupt Controller (VIC) with up to 16 priority levels to interrupt the ARM processor on request. The software development is facilitated by the standard ARM Test/Debug interface with Real-Time In-Circuit Emulator.



SJA2010 block diagram

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com). A complete list will be sent to you automatically. You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2005

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

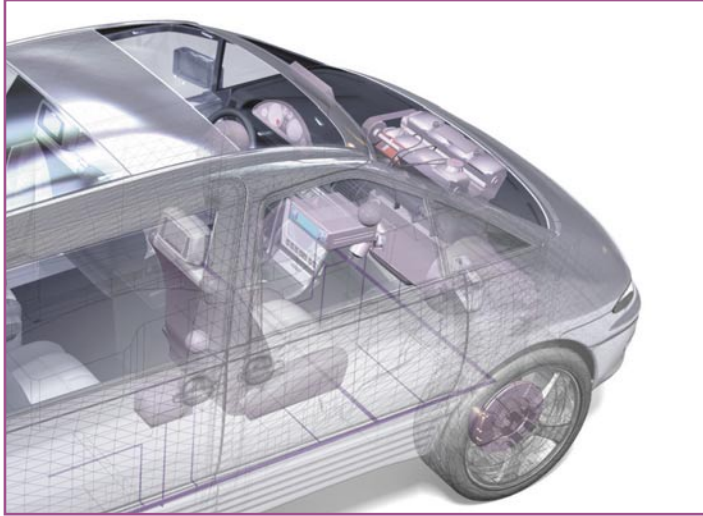


Date of release: February 2005  
Document order number: 9397 750 14654

Published in The Netherlands

# AU578x J1850 VPW transceiver family

Our dedicated network devices include a range of fully compatible J1850 transceivers, suited to mid-speed non real-time communications and diagnostics for in-vehicle applications.



## Key features

- Supports SAE/J1850 VPW standard for in-vehicle class B multiplexing
- Bus speed 10.4 kbits/s nominal
- Drive capability 32 bus nodes
- Thermal overload protection
- Diagnostic loop-back mode
- Very low sleep mode power consumption
- 4X mode (41.6 kbits/s) data reception capability

## AU5780A

- Low RFI due to output wave shaping with adjustable slew rate
- Direct battery operation with protection against +50V load dump, jump start and reverse battery connection
- Bus terminals proof against automotive transients up to -200V/+200V
- ESD protected to 9 kV on bus and battery pins
- 8-pin SO package

## AU5783

- Supports 41.6 kbits/s for data upload and download
- Low RFI due to output wave shape function
- Direct battery operation with protection against +40V load dump and 8 kV ESD
- Bus terminals proof against automotive transients up to +100V/-150V and 8 kV ESD
- Power supply enable function
- 14-pin SO package

## Supporting the SAE/J1850 standard for low-cost networking

### Semiconductors

The SAE/J1850 standard establishes the requirements for a class B data communication network interface for all vehicles. It defines a minimum set of data communication requirements for a network that is cost-effective for simple applications yet flexible enough to be used in complex applications. In the USA, SAE/J1850 has been adopted as a non real-time communications and diagnostics bus, and Philips Semiconductors' portfolio of automotive J1850 transceivers underlines the company's support for the standard.

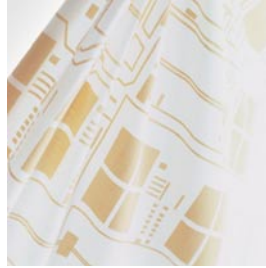
The AU5780/83 family of line transceivers is intended primarily for in-vehicle multiplex applications. It provides interfacing between a link controller and the physical bus wire, at the nominal bus speed of 10.4 kbits/s with a 4X transmission mode supporting data upload and download at a nominal speed of 41.6 kbits/s. A sleep mode keeps power consumption low.

The AU5780A has protection against higher load dumps and transient signal spikes than the AU5783. However, the AU5783 also provides protection against loss of ground conditions, ensuring the network will continue to operate in case of an electronic control unit losing connection to ground potential. In addition, the AU5783 can turn off an external voltage regulator, further reducing overall power consumption. The voltage regulator is re-activated upon detection of bus activity or a local wake-up event.

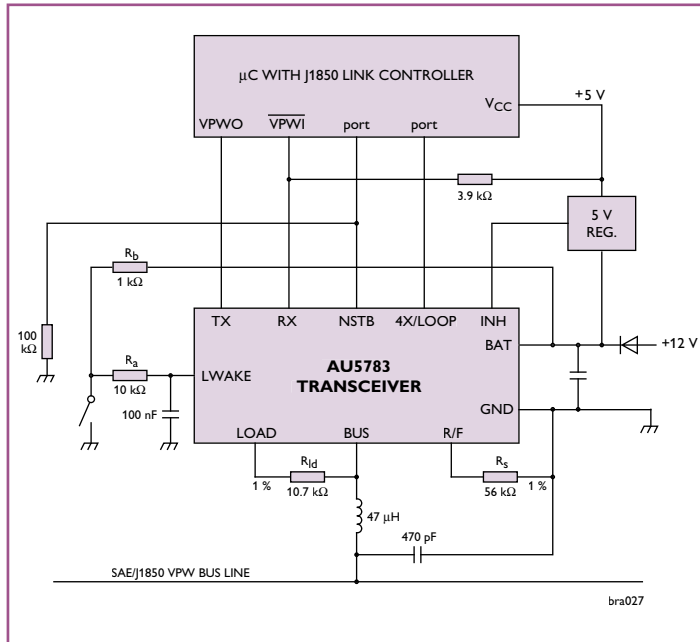
# PHILIPS

# AU578x J1850 VPW transceiver family

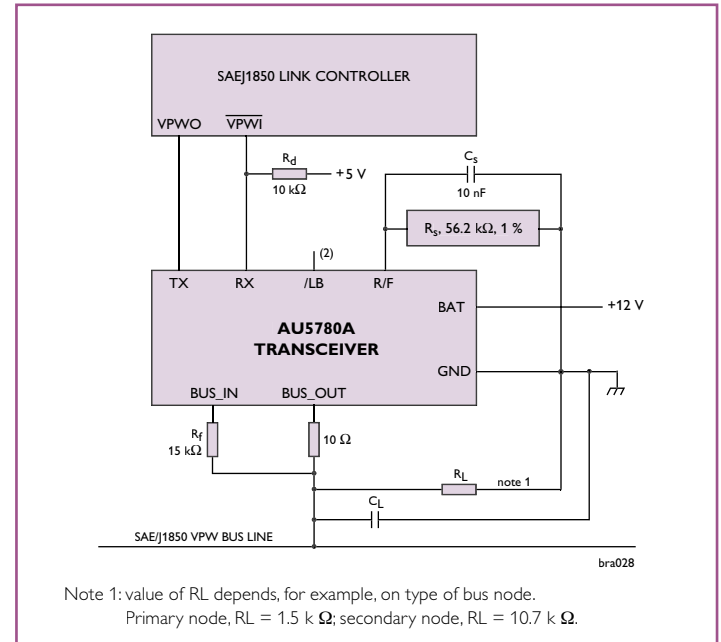
Supporting the SAE/J1850 standard for low-cost networking



[www.semiconductors.philips.com](http://www.semiconductors.philips.com)



AU5783 block diagram



AU5780A block diagram

## Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.

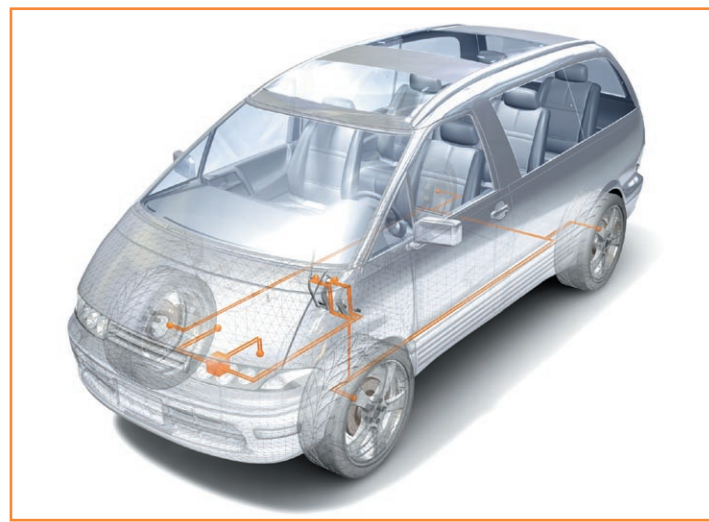


Date of release: June 2004  
Document order number: 9397 750 13205

Published in The Netherlands

# FlexRay: X-by-wire networking

Supporting the shift from mechanical to electrical in-car systems, Philips is working as a FlexRay Consortium Core Partner to develop a complete X-by-wire communications system, starting with the physical layer, capable of supporting the in-vehicle networking systems of tomorrow.



## Key features

### Physical layer

- Data transfer up to 10 Mbit/s
- Deterministic data transmission, guaranteed message latency and message jitter
- 'Bus Guardian' ensures fault containment in the time domain
- Flexible and extensible network layout: support for bus, star and multiple star topologies
- Robust network wake-up mechanism
- Advanced power mode handling
- Fault separation in active star networks

### Protocol

- Fault-tolerant and time-triggered services implemented in hardware (e.g. synchronized global time base)
- Support of repetitive as well as spontaneous message scheduling
- Dedicated online diagnosis services
- Support of network management services and efficient message filtering mechanisms
- Support of redundant transmission channels
- Robust coding and bit recognition scheme

# The complete intelligent in-car networking system

## Semiconductors

A new car is among the most expensive purchases a consumer will make – and safety, price and economy are often buyers' primary concerns. This impels manufacturers to find ways of making their cars more efficient, easier to drive, better protected in an accident and more reliable. In many cases this means switching from bulky, heavy mechanical systems to lightweight electrical systems, significantly increasing the electronic content in the average car – a trend that's set to continue.

The generic term 'X-by-wire' covers all the technologies needed to interconnect automotive control applications, from steering to braking and acceleration, supplementing and ultimately replacing current mechanical and hydraulic solutions. As a core technology provider to the FlexRay Consortium, Philips is working hand-in-hand with car manufacturers DaimlerChrysler, BMW, GM and Volkswagen as well as many other well-known automotive suppliers such as Bosch and Motorola, to integrate the many benefits of X-by-wire into our future cars.

The FlexRay Consortium is developing the FlexRay communications protocol and physical layer – the core of a system supporting future in-car high-speed control applications. FlexRay addresses the demand of increased bandwidth to handle the growing number of sensors, actuators and control systems for the various safety, reliability and comfort systems within vehicles. Providing a common, scalable communications system with the potential for expansion in the future is a key goal for FlexRay. Alongside the protocol, software and support services in development, the FlexRay Consortium also aims to ensure the provision of tools for design, measurement and simulation of the communications system through industry-leading tool-makers and test houses grouped in the Consortium's Development Members community. Philips is currently developing silicon solutions for the FlexRay system.

# PHILIPS



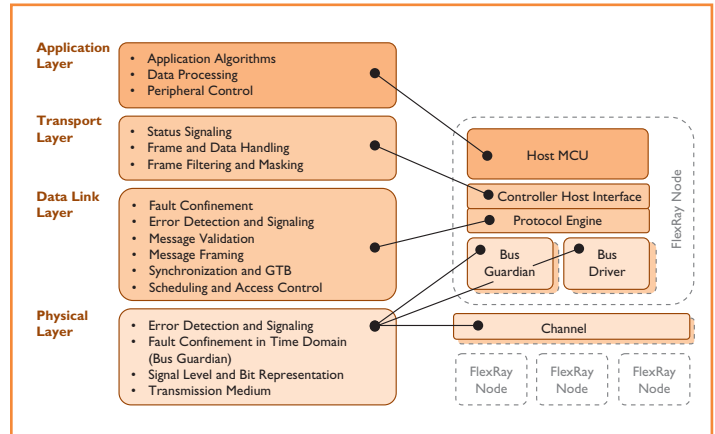
# FlexRay: X-by-wire networking

## The complete intelligent in-car networking system



This includes bus drivers and bus guardians, for which prototype samples have already undergone both laboratory and in-field testing, with promising results. Based on the feedback obtained, device specifications are currently being refined and future production bus drivers and bus guardians are in development, with the TJA1080 FlexRay Bus Driver being the first to be released from the roadmap.

The Consortium is driving FlexRay forward as the automotive industry standard for 'X-by-Wire.' And as a FlexRay Core Partner, Philips could not be better positioned to develop the silicon solutions for this evolving standard – bringing drivers a safer and more comfortable ride.



FlexRay Node Architecture

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



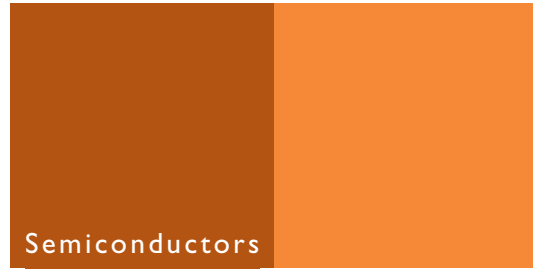
Date of release: June 2004  
Document order number: 9397 750 12202

Published in The Netherlands

# TJA1080 FlexRay Bus Driver

As Philips' first silicon solution for the FlexRay electrical physical layer, the TJA1080 high-speed time-triggered communication system transceiver chip highlights FlexRay's power and versatility. Offering an extensive feature set, excellent EMC performance and unique possibility to be used also in active star couplers the TJA1080 is helping drive this new advanced communications system into tomorrow's cars.

## High-speed time-triggered communication system transceiver chip



Philips' high-speed time-triggered communication system transceiver chip offers an extensive feature set, excellent EMC performance and unique potential for use in active star couplers. Combining FlexRay transceiver and FlexRay active star device on a single chip, the TJA1080 supports the full bandwidth of FlexRay, from 500 kbit/s to 10 Mbit/s. Internal voltage and temperature monitoring, bus error detection and a safety time-out make the TJA1080 a cornerstone of any FlexRay ECU.

Dedicated error and status information – readable by any microcontroller – allows precise monitoring of system performance. In active star and node configurations, features such as Sleep and Standby modes serve the need for automotive power management. A Bus Guardian interface rounds out the feature list and enables the TJA1080 to be used in systems that require additional supervisions.

### Key features

- Data transfer up to 10 Mbits/s
- Compliant with FlexRay electrical physical layer specification
- Fail-silent behavior via Bus Guardian control input
- Interface to optional Bus Guardian available for bus access supervision
- Excellent EMC performance (due to 3rd generation Silicon-On-Insulator process)
- High common mode range ensures excellent EMI
- 12, 24 and 42 V system support with low sleep current (maximum 50  $\mu$ A)
- Supports 2.5, 3.0, 3.3 and 5 V microcontrollers and automatically adapts to interface levels
- Integrated automotive power management system
  - two INH-pins enabling power consumption optimized ECU design
  - local wake-up input
  - remote wake-up capability via FlexRay bus in low power modes
  - autonomous power modes, when used in active stars
- $\pm 8$  kV ESD protection (HBM) on BP and BM pins
- Automotive temperature range  $-40$  °C to  $+125$  °C
- Small footprint SSOP20 package (5.4 mm x 7.4 mm body)

### Key benefits

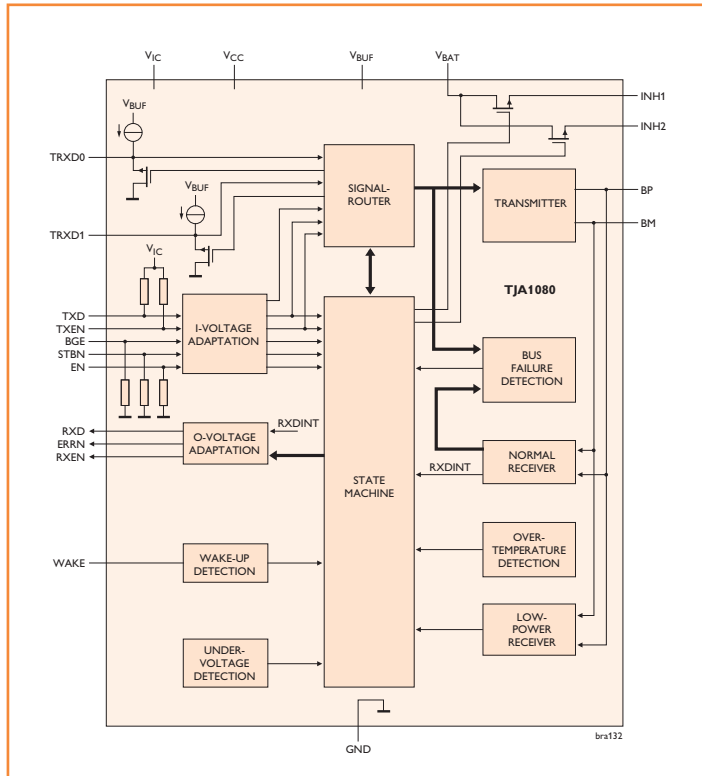
- Supports wide range of microcontrollers, maximizing flexibility and lowering cost of nodes
- Enables autonomous active star couplers without a microcontroller
- Identical mode control as for TJA1041(A) and TJA1054(A) CAN transceivers
- Simplifies development of robust and fully fail-silent communication nodes



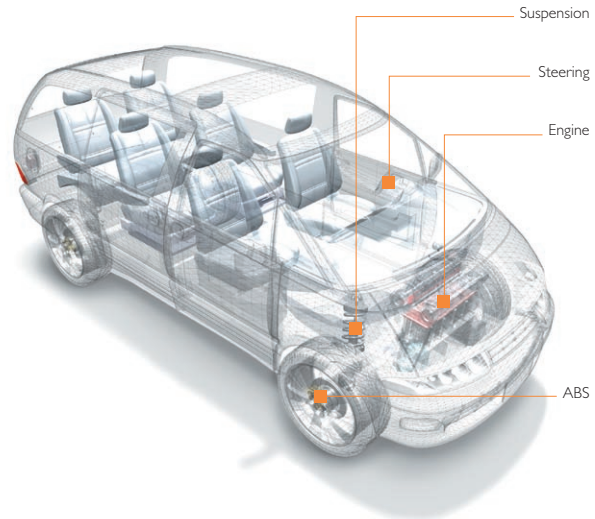
# PHILIPS

# TJA1080 FlexRay Bus Driver

High-speed time-triggered communication system transceiver chip



TJA1080 block diagram



## Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com).

A complete list will be sent to you automatically.

You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



Date of release: June 2004  
Document order number: 9397 750 13048

Published in The Netherlands

# Safe-by-Wire Plus™

## Connectivity to ensure occupant safety

The Safe-by-Wire Plus consortium has defined a global networking standard for automotive occupant safety systems that merges the best features from existing Safe-by-Wire and BST developments. This new industry standard will benefit both automotive manufacturers and consumers by easing the integration of advanced safety features, leading to safer vehicles throughout the price spectrum.



### Key features

- Two-wire bus system for sensor and deployment networks
  - Bi-directional master-slave communication
  - Slave devices powered by master via bus
- Flexible bus topology
  - Supports bus, tree, ring and daisy-chain topologies
- Highly efficient protocol with minimum overhead
  - Selective deployment of up to 12 squibs within 200  $\mu$ s
  - Data acquisition for 3 sensors (8 bits each) within 250  $\mu$ s (4 kHz)
  - Flexible diagnostic communication
- Fault-tolerant
  - Real-time recovery from bus wire shorts
  - Individual bit error checking using CRC
  - Data validation by transmitter guards against bus collision errors
  - Independent 'safety' signal for deployable devices
- Variable bus speed (20, 40, 80 or 160 kbits/s)
  - Can even be changed 'on-the-fly'
  - Slaves synchronize easily to bus speed given by master
- Supports various sensor update configurations, including:
  - A mix of 4 kHz and 2 kHz sensors on the same bus
  - A mix of fast (2 kHz / 4 kHz) impact sensors and slow (< 1 kHz) occupancy sensors on the same bus

To deliver the highest possible safety standards, automotive manufacturers are incorporating more and more safety features across their model ranges. Seat-belt pre-tensioners, multiple airbags and an array of sensors (impact, acceleration, pressure, occupancy, roll-over, etc.) are being combined into adaptive occupant restraint systems. Ensuring the right safety feature is activated at the right time requires quick, reliable communication between the restraints control module and all these sensors plus a host of deployment devices. The members of the Safe-by-Wire Plus consortium are leveraging their extensive combined knowledge and experience to refine the global networking standard for these communications that will bring the highest safety levels to all vehicles.

The Safe-by-Wire Plus consortium comprises leading automotive systems and component suppliers, uniting the previous activities of the Safe-by-Wire consortium (Analog Devices Inc., Autoliv Inc., Delphi Corporation, Key Safety Systems, Philips, Special Devices Inc. and TRW Automotive) and the BST group (Bosch, Continental Temic and Siemens VDO Automotive). The key goal was to combine the independent and competing automotive safety bus specifications into a single global standard, capturing the best of both previous implementations. Through component standardization, the new consortium expects to reduce development costs and enable automotive manufacturers to leverage the economies of scale – allowing more efficient integration of advanced safety features. This will ultimately contribute to safer vehicles at all price levels.

The consortium has recently released version 2.0 of the Automotive Safety Restraints Bus (ASRB 2.0) specification. Based on the Safe-by-Wire ASRB 1.0 specification, it also includes some of the concepts from the work conducted by the BST group. The new specification is fully compliant with the Safe-by-Wire physical layer, preserving compatibility with Safe-by-Wire (ASRB 1.0) systems and components already being developed. The consortium has submitted the ASRB 2.0 specification to the appropriate ISO working group for consideration as a global standard (ISO 22896). Furthermore, the Safe-by-Wire Plus solution is a royalty-free standard.

# PHILIPS

# Safe-by-Wire Plus™

## A complete portfolio



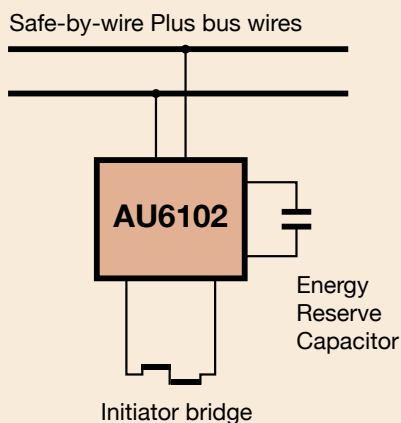
Semiconductors

Philips continues to develop a complete set of Safe-by-Wire / Safe-by-Wire Plus interface ICs for restraint control modules and sensor / deployment slaves. Designed for use in today's advanced automotive safety systems, these devices have the flexibility to accommodate tomorrow's innovations. The portfolio is fully compliant with the Safe-by-Wire Plus defined standard, simplifying the design of passenger restraint systems, allowing rapid customization and reducing costs so that the highest protection levels are available in all vehicles.

### AU6102 – deployment slave

The AU6102 is an engineering prototype version of a single-chip parallel bus interface for pyrotechnical low-energy igniters. This squib driver includes a charge pump for storage of the firing energy in an external capacitor and provides full diagnostics for the initiator bridge.

- Safe deploy control by two independent deploy switches
- Energy reserve control
- Diagnostics for deploy switches and initiator bridge
- OTP memory for address and traceability data





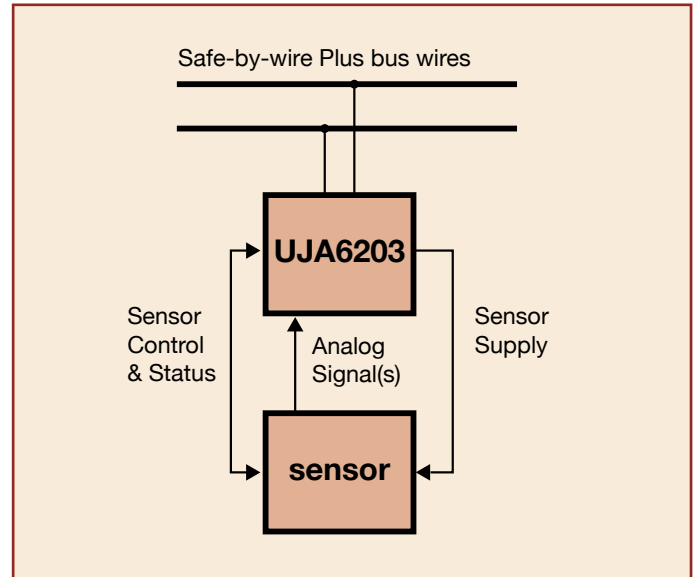
# Safe-by-Wire Plus™

## A complete portfolio

### UJA6203 – raw-data sensor slave

The UJA6203 is a single-chip parallel sensor interface capable of providing communication between an analog impact sensor and the Safe-by-Wire Plus bus. Analog inputs from one or two crash sensors are received by the UJA6203, converted into a Safe-by-Wire Plus compatible digital format and sent as raw data over the bus. The device itself is self-powered, but it has a built-in 5V regulator capable of supplying external loads.

- Raw data Safe-by-Wire Plus sensor bus interface
- Built-in 10-bit 2-channel ADC
- Powered directly from bus
- Regulated 5V output
- Built-in diagnostic capabilities and supports sensor health monitor
- OTP memory for address, set-up and traceability data
- SO8 package

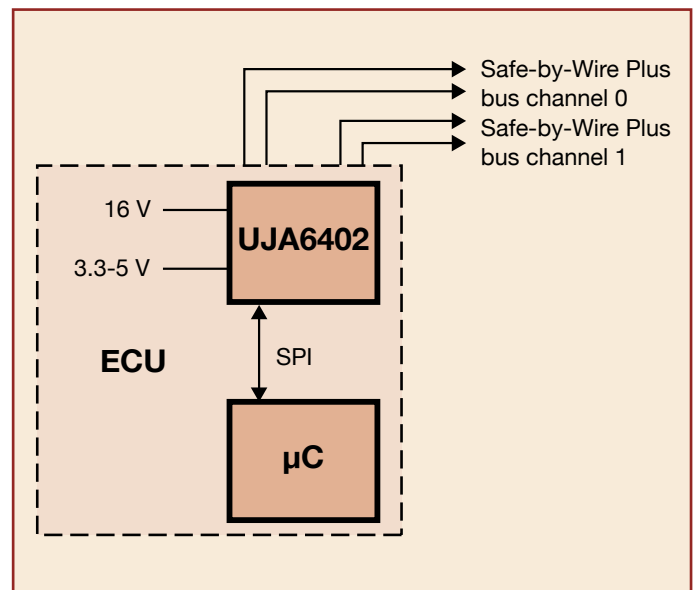


Raw data sensor interface

### UJA6402 - dual channel sensor bus master

Philips' UJA6402 dual-channel sensor bus master provides the interface between two Safe-by-Wire Plus busses and a vehicle's main occupant safety ECU. The master communicates with the Safe-by-Wire Plus sensor slaves via the bus and is controlled via the SPI port by a local ECU microcontroller. A fully self-contained solution, the UJA6402 incorporates Safe-by-Wire Plus data link controller and transceivers along with the required floating supplies and diagnostic capabilities.

- Drives two independent Safe-by-Wire Plus buses
- Supports sensor bus slaves
- SPI port for direct microcontroller interface, 3.3 – 5V
- Fault-tolerant
- Allows individual data length selection for each S-frame slot (8 or 10 bit)
- Choice of 3- or 8-bit CRC for S-frames
- HTSSOP32 package
- EMC compliant



ECU interface

# Safe-by-Wire Plus™

## Connectivity to ensure occupant safety

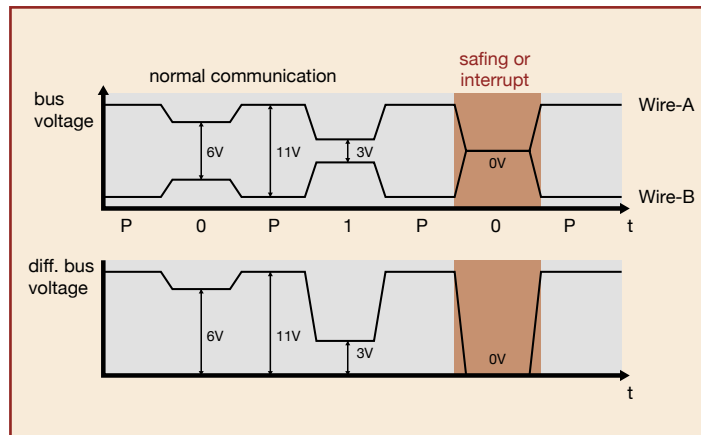


### System advantages

Replacing conventional point-to-point connections with a Safe-by-Wire Plus bus brings numerous benefits to your system, that reduce overall cost and increase flexibility.

- Simplified wiring and cost-effective approach
  - Shorter cabling reduces weight
  - Smaller ECU connector saves space
  - Re-use of standardized components cuts development time
  - Standardization leads to high-volume multi-source components
- Future-proof open bus architecture simplifies system changes
  - Modify existing slaves
  - Use slaves from different sources
  - Change the number of slaves
  - Add new types of slaves
- Electronic deploy interface supports safe system design
  - Airbag deployment only by bus message makes handling safer
  - Individual, local deployment energy per airbag can be de-powered for airbag suppression
  - Electronic identification of slaves allows check of system integrity
  - Tamper resistant

The Safe-by-Wire Plus specification can be downloaded from  
<http://www.semiconductors.philips.com/automotive/ivn>



Absolute and differential signal

### Philips Semiconductors

Philips Semiconductors is a worldwide company with over 100 sales offices in more than 50 countries. For a complete up-to-date list of our sales offices please e-mail [sales.addresses@www.semiconductors.philips.com](mailto:sales.addresses@www.semiconductors.philips.com). A complete list will be sent to you automatically. You can also visit our website <http://www.semiconductors.philips.com/sales>.

© Koninklijke Philips Electronics N.V. 2004

All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent- or other industrial or intellectual property rights.



Date of release: November 2004  
Document order number: 9397 750 14332

**Published in The Netherlands**